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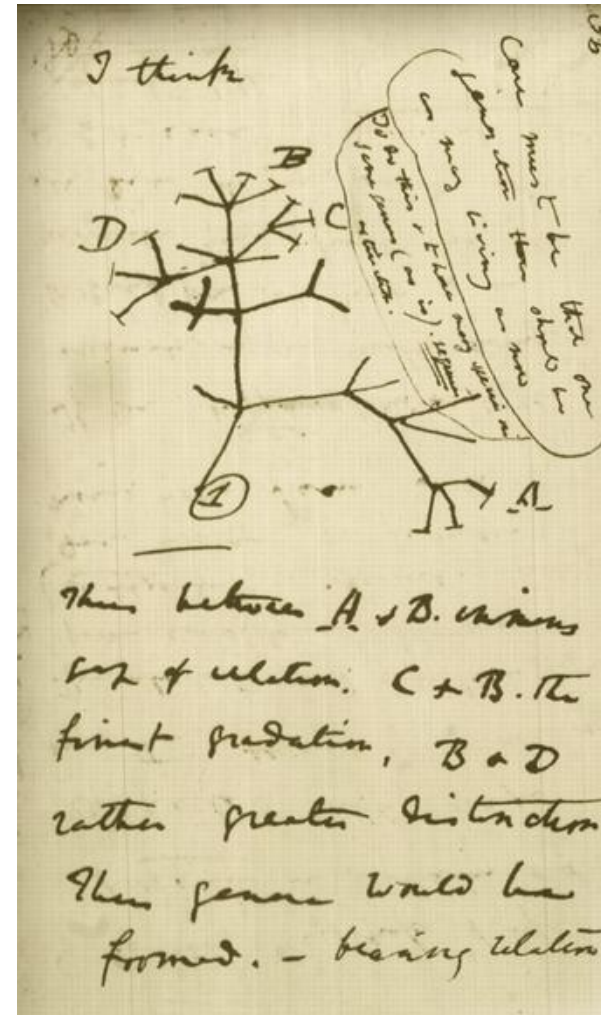
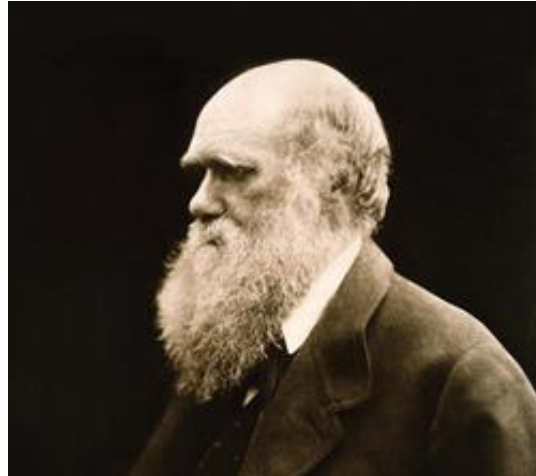
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Introduction to Phylogenetics

Marcela Suárez-Esquivel

Bacterial Genomic Surveillance
June 2026

Darwin's tree of life



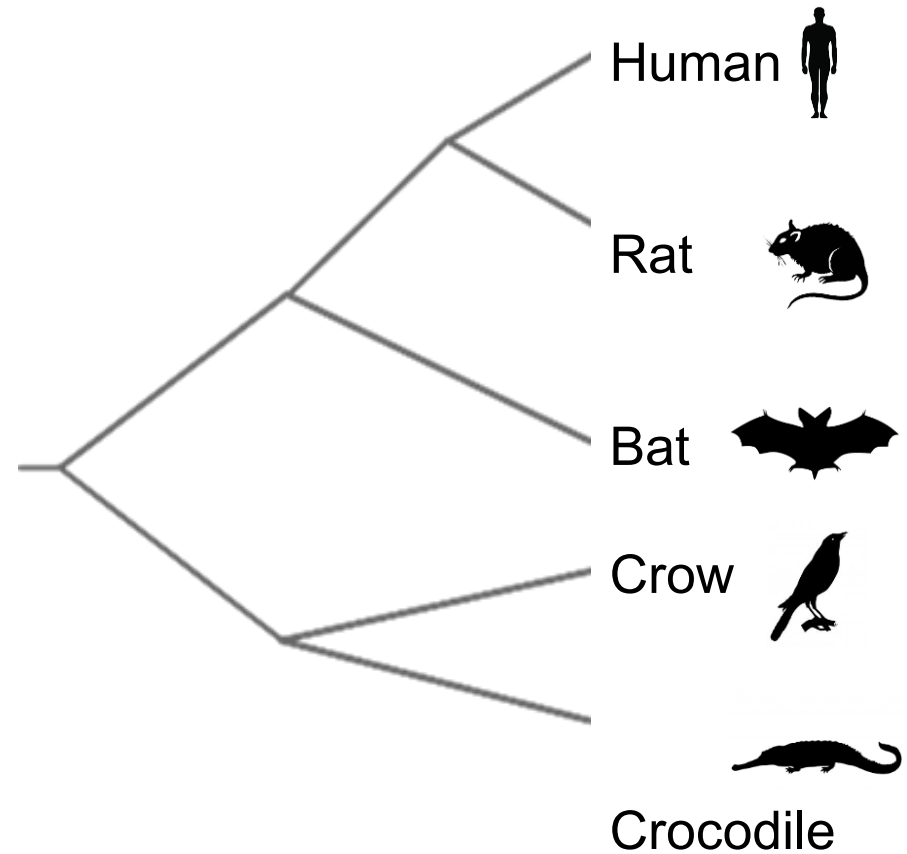
Single Nucleotide Polymorphisms (SNPs)

Seq1 GTAAAATAGGCACCATCAAAACGCAGAGGGGAAGACGGG**ATG**
Seq2 GCAAAAATAGGCGCCACAGAAACGCAGAGGGGAAGACGGGA**ATG**
Seq3 GCAAAAATAGGCGCCTCCGAAACGCAGAGGGGAAGACGGG**ATG**

- Organisms accumulate genetic variation through natural processes, e.g. drift
- This can be seen when comparing a group of DNA sequences

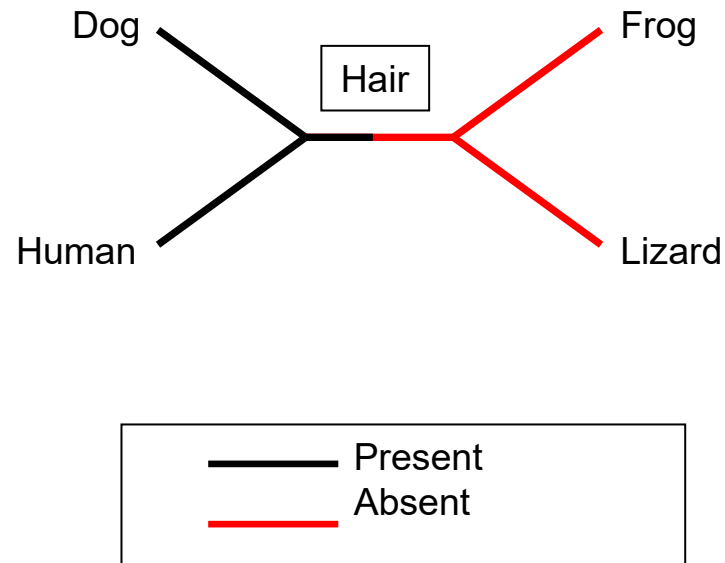
Phylogenetic Systematics

- **Phylogenetics** is the study of the evolutionary relationships among homologous items (taxa)
- A **phylogeny** is a hypothetical model of the evolutionary relationships shared among these taxa and is usually portrayed as a dendrogram (tree)
- Trees can represent:
 - 1- The evolutionary history of organisms/individuals (species tree)
 - 2- The evolutionary history of a gene, group of genes (gene tree), or whole genome.
- To build trees we can use rRNA sequences, concatenated gene sequences, whole genomes, SNPs

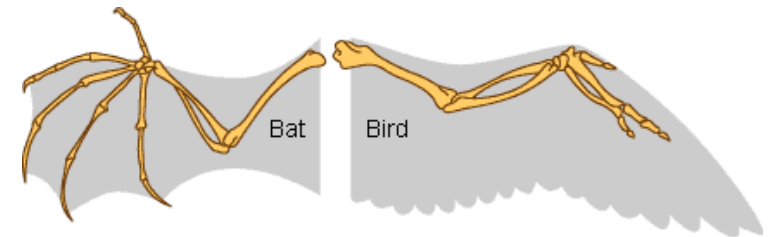
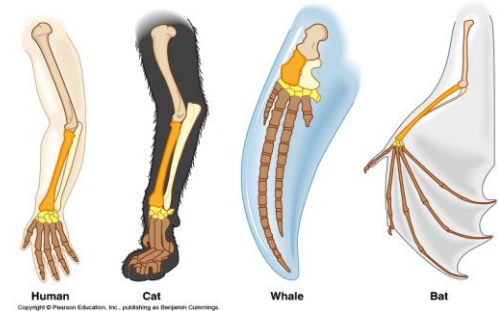


Similarity, homology, homoplasy, analogy

Homology: Similarity is due to common inheritance from a common ancestor.



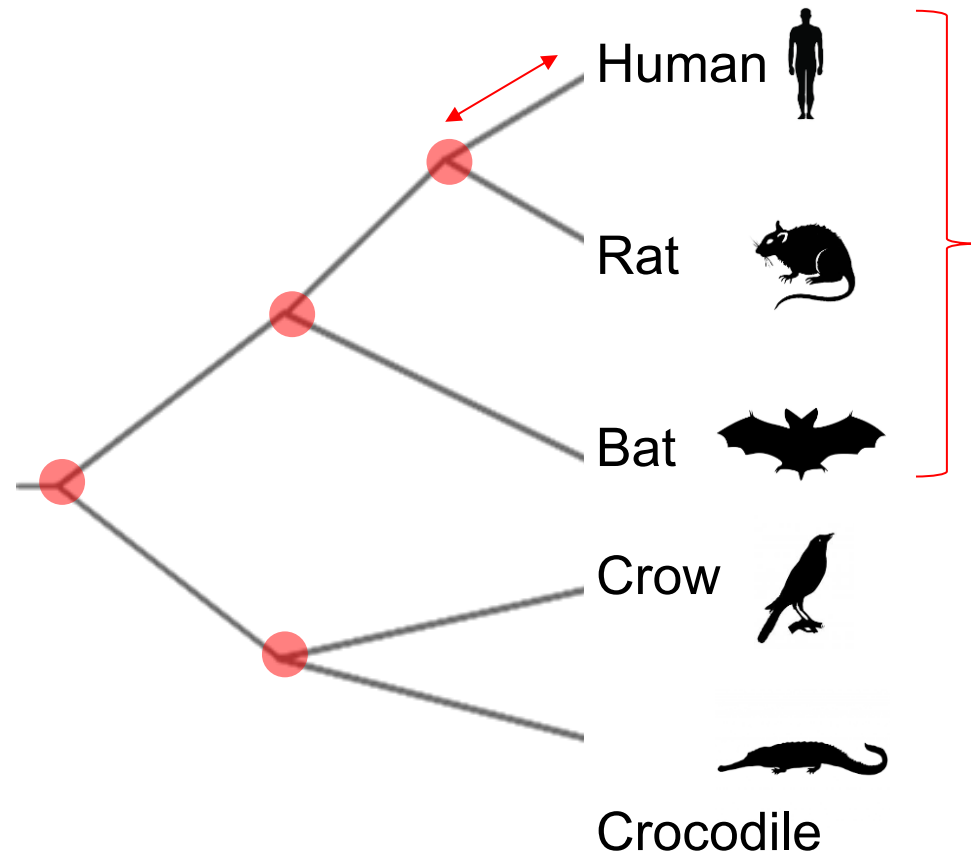
Homoplasy: Similarity is due to convergent evolution. The character was absent from the nearest common ancestor of the two taxa. Independent acquisitions of the same or superficially similar character state. Homoplastic characters provide a misleading picture of phylogeny.



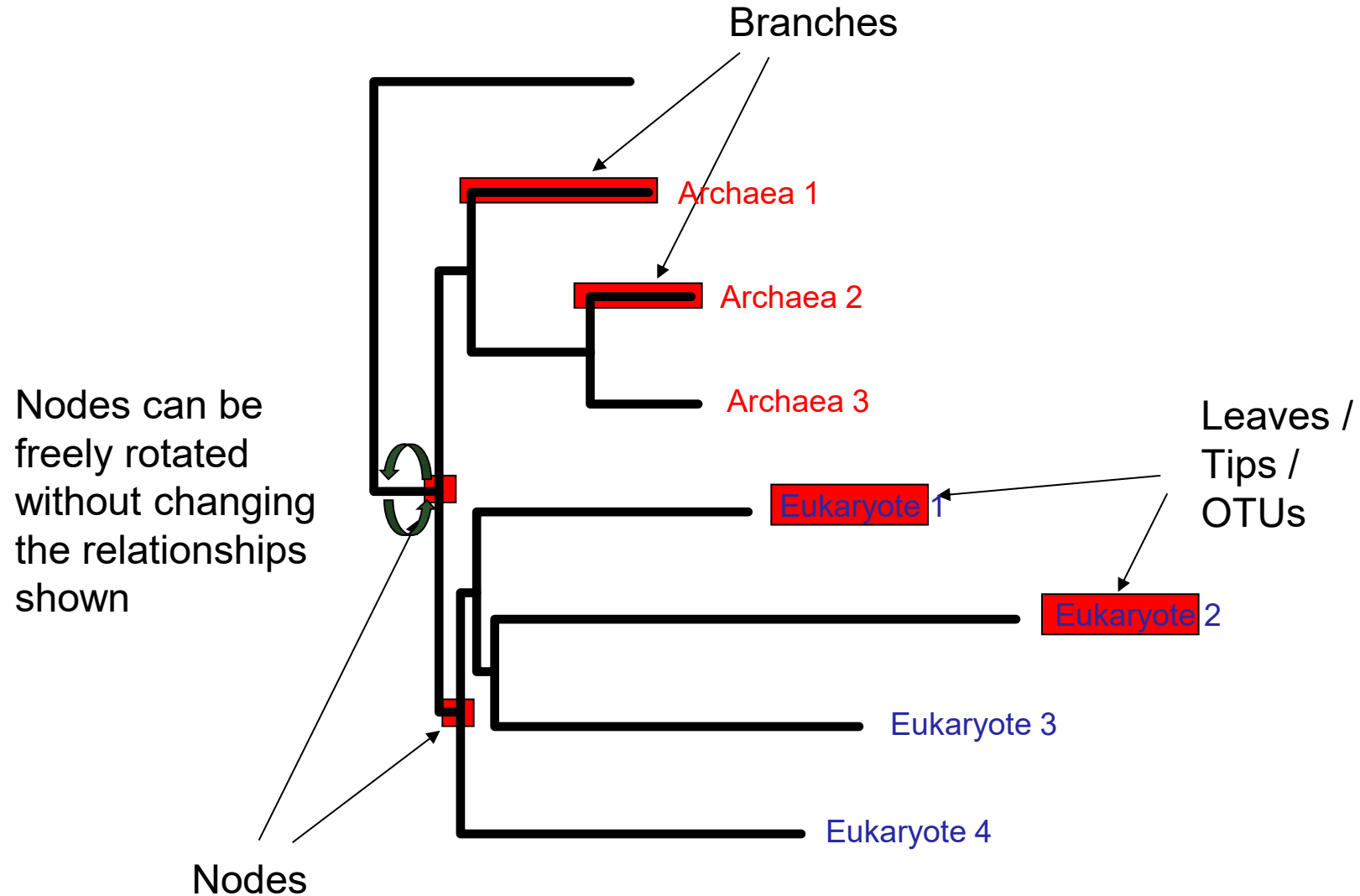
Anatomy of a Phylogenetic tree

- Taxa represent sequences, e.g. extant organisms (species alive today) or bacterial isolates
- **Nodes** represent the nearest common ancestor between the taxa it encompasses (or the sequences present in said ancestor).
- Distance between taxa reflects a decreasing number of shared, homologous characters

- Monophyletic groups (clades) contain taxa that are more closely related to each other than to any taxa outside the group

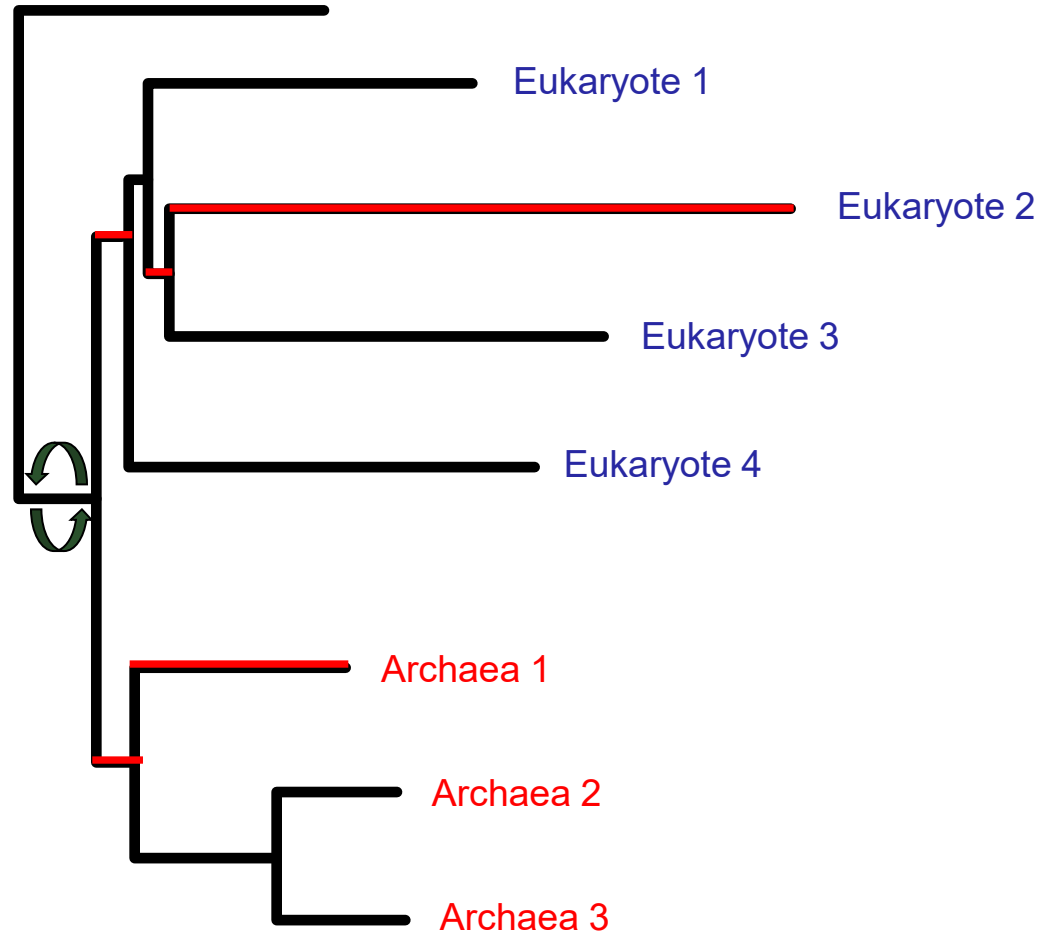



Some Tree Terms and Facts



Some Tree Terms and Facts

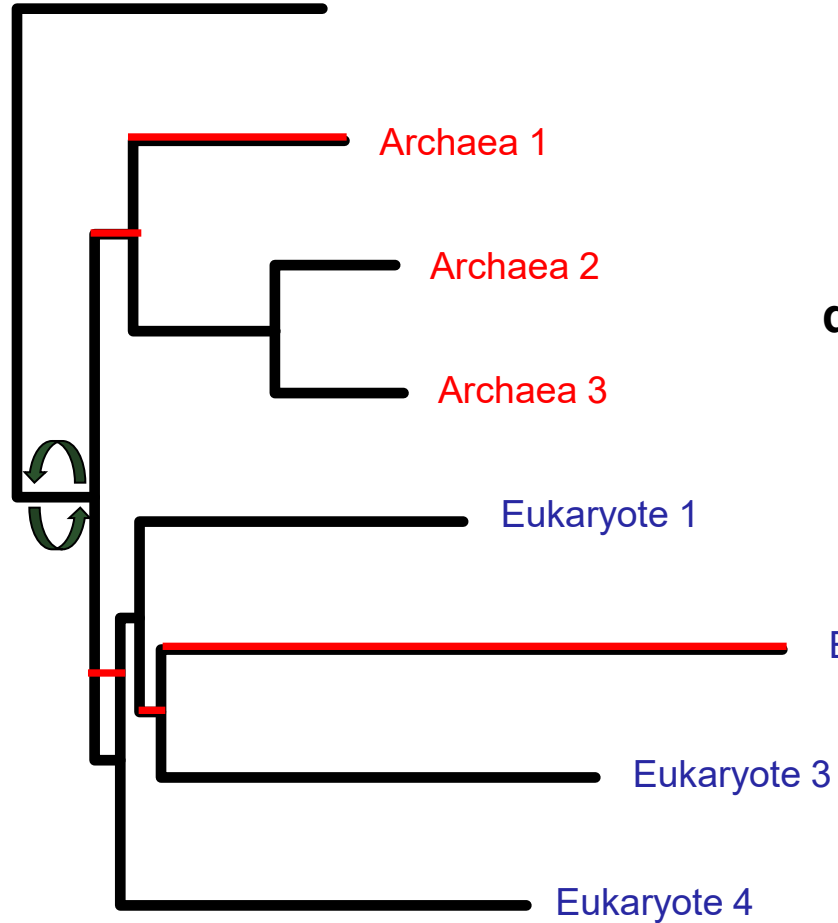
Nodes can be
freely rotated
without changing
the relationships
shown



Total distance = 

Some Tree Terms and Facts

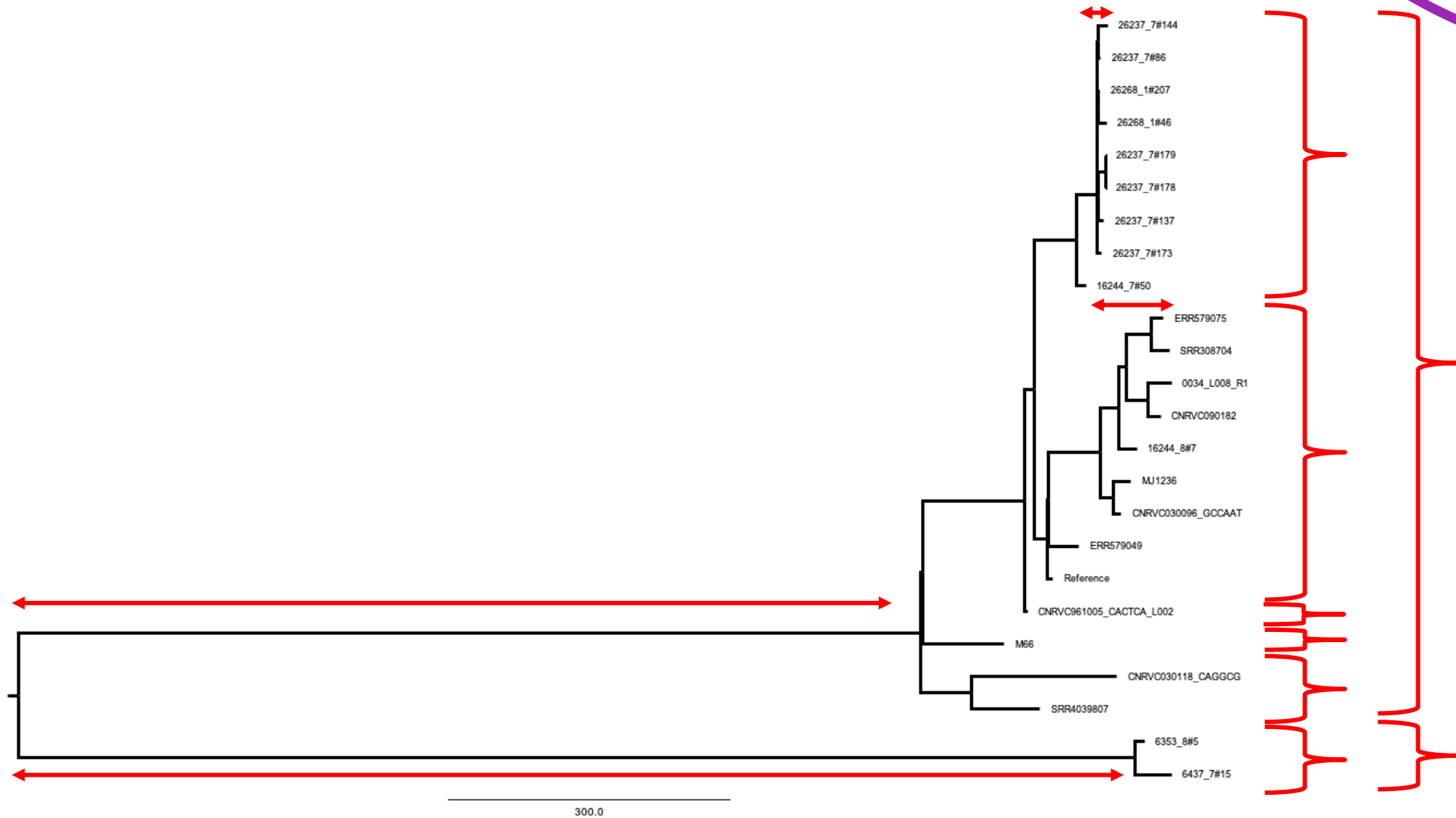
Nodes can be
freely rotated
without changing
the relationships
shown



**Only horizontal
distances indicate
divergence**

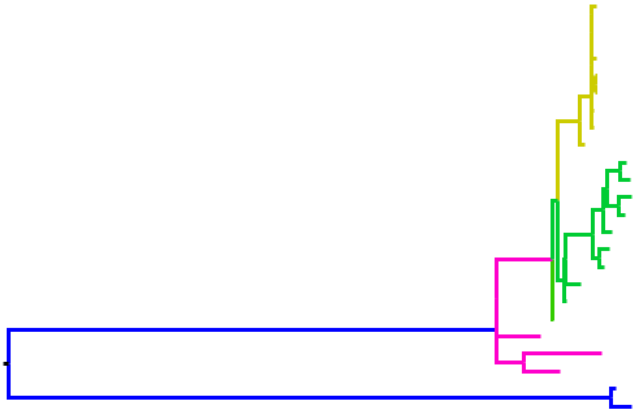
Total distance =

Interpreting a Phylogenetic Tree

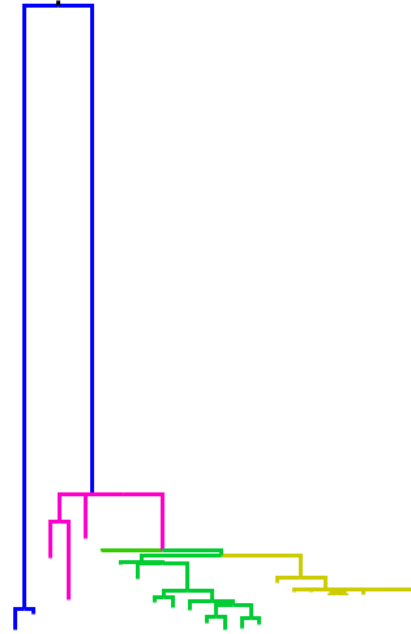


The same tree can be displayed in different ways

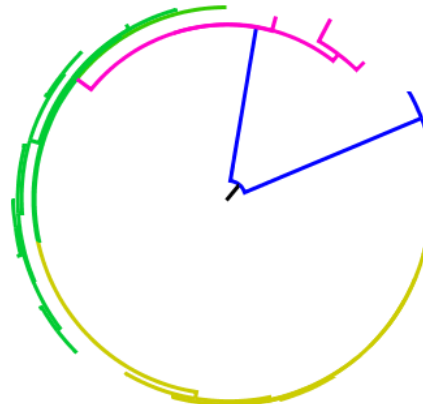
Linear (horizontal)



Linear (vertical)

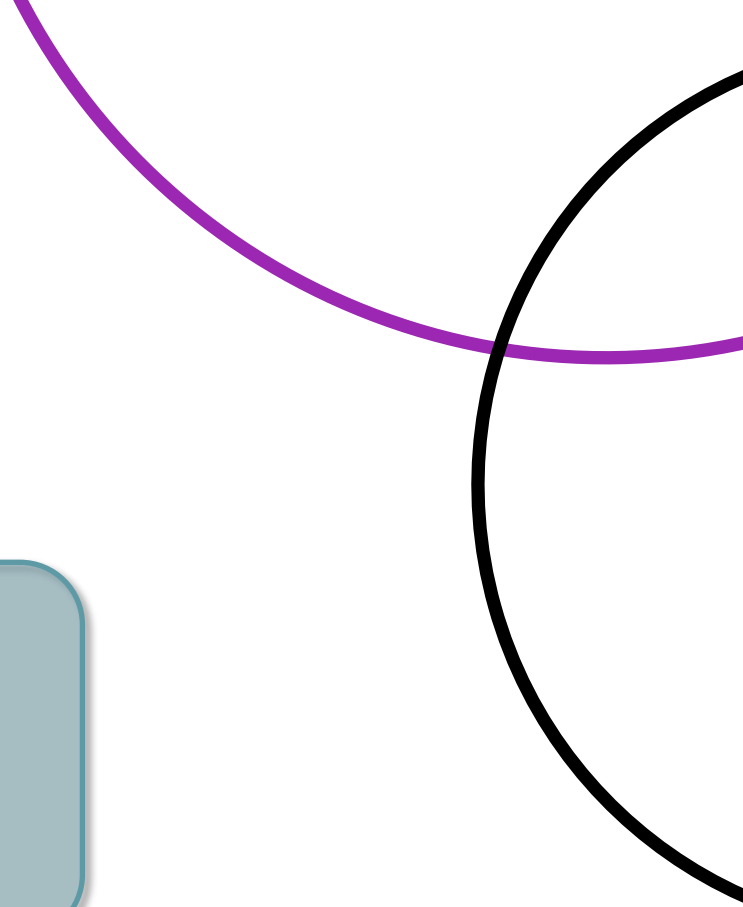


Circular



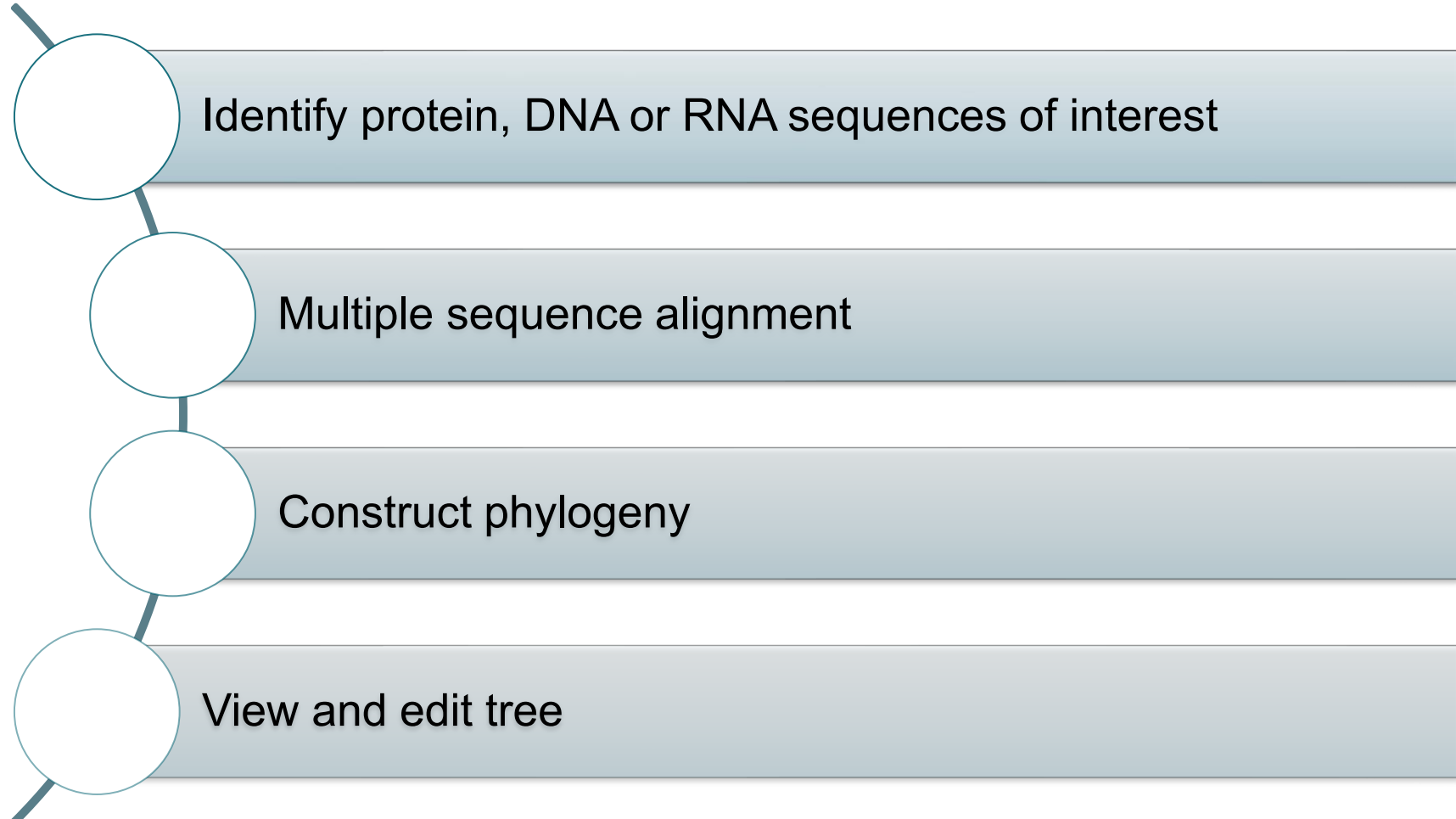
Unrooted





How do we build a
phylogenetic tree?

Building a Phylogenetic Tree

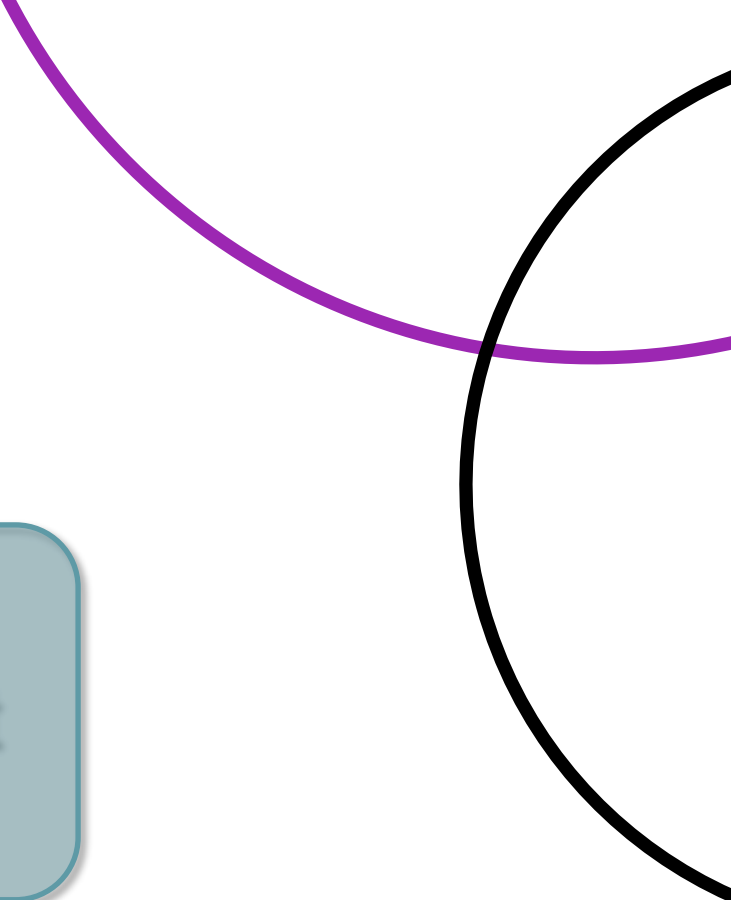


Identify protein, DNA or RNA sequences of interest

Multiple sequence alignment

Construct phylogeny

View and edit tree



Multiple sequence alignment

Multiple sequence alignment

A multiple sequence alignment is your best hypothesis of **positional homology** between bases/amino acids of different sequences

```
sel=0 71
ERS009884 SC contig0 GDVIGKYHPHGDSAVYDTIVRMAQ
ERS009906 SC contig0 GDVIGKYHPHGDLAVYDTIVRMAQ
ERS009795 SC contig0 GDVIGKYHPHGDSAVYDTIVRMAQ
ERS009819 SC contig0 GDVIGKYHPHGDSAVYDTIVRMAQ
ERS009835 SC contig0 GDVIGKYHPHGDSAVYDTIVRMAQ
ERS009842 SC contig0 GDVIGKYHPHGDSAVYDTIVRMAQ
ERS009843 SC contig0 GDVIGKYHPHGDSAVYDTIVRMAQ
ERS009858 SC contig0 GDVIGKYHPHGDSAVYDTIVRMAQ
ERS009869 SC contig0 GDVIGKYHPHGDSAVYDTIVRMAQ
ERS008193 SC contig0 GDVIGKYHPHGDLAVYDTIVRMAQ
```

If your sites are not homologous, it will introduce error into your phylogeny

Alignment can be easy or difficult

GCGGCCCA	TCAGGTAGTT	GGTGG
GCGGCCCA	TCAGGTAGTT	GGTGG
GCGTCCA	TCAGCTGGTT	GGTGG
GCGTCCA	TCAGCTAGTT	GGTGG
GCGGCACA	TAGCTAGTT	GGTGA
*****	*****	*****

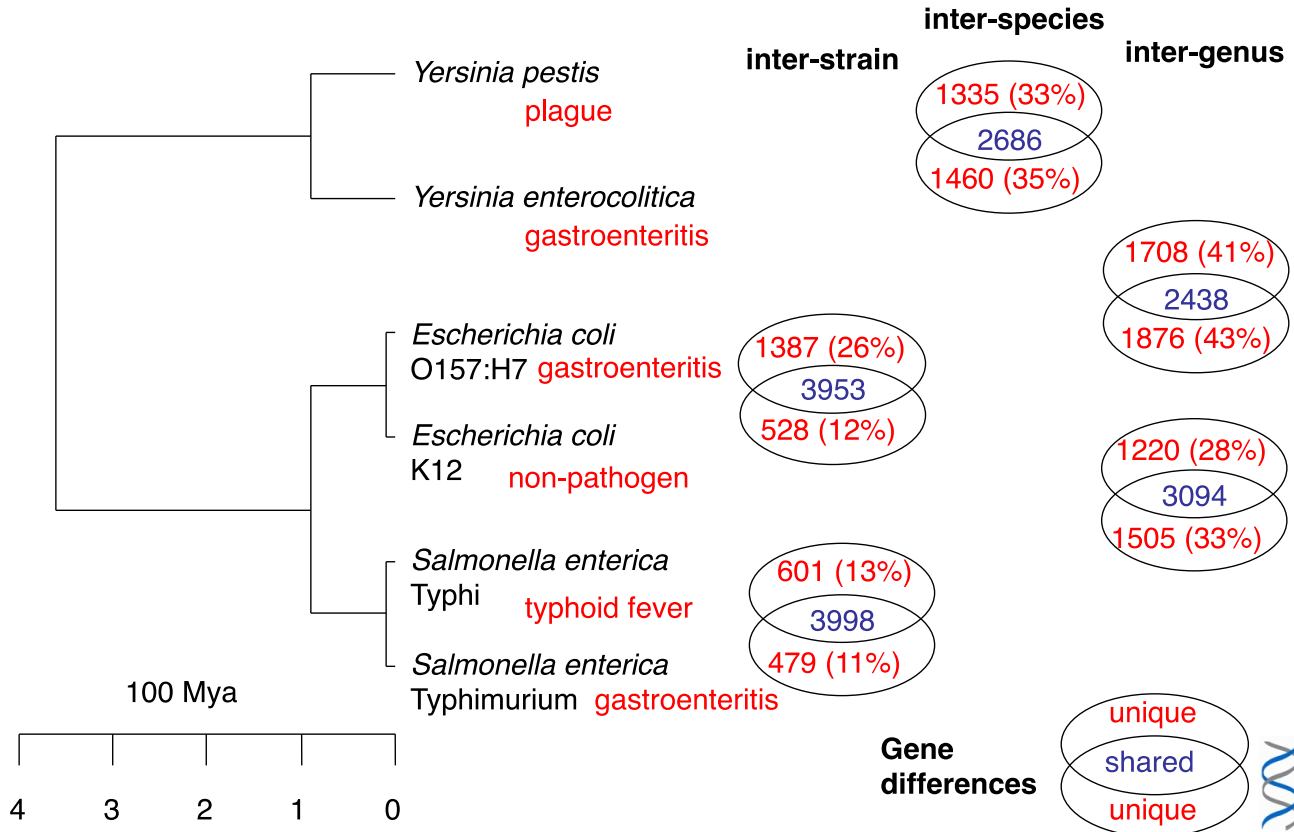
Easy

TTGACATG	CCGGGG---A	AACCG
TTGACATG	CCGGTG--GT	AAGCC
TTGACATG	-CTAGG---A	ACGCG
TTGACATG	-CTAGGGAAC	ACGCG
TTGACATC	-CTCTG---A	ACGCG
*****	??????????	*****

Difficult due
to insertions
or deletions
(indels)

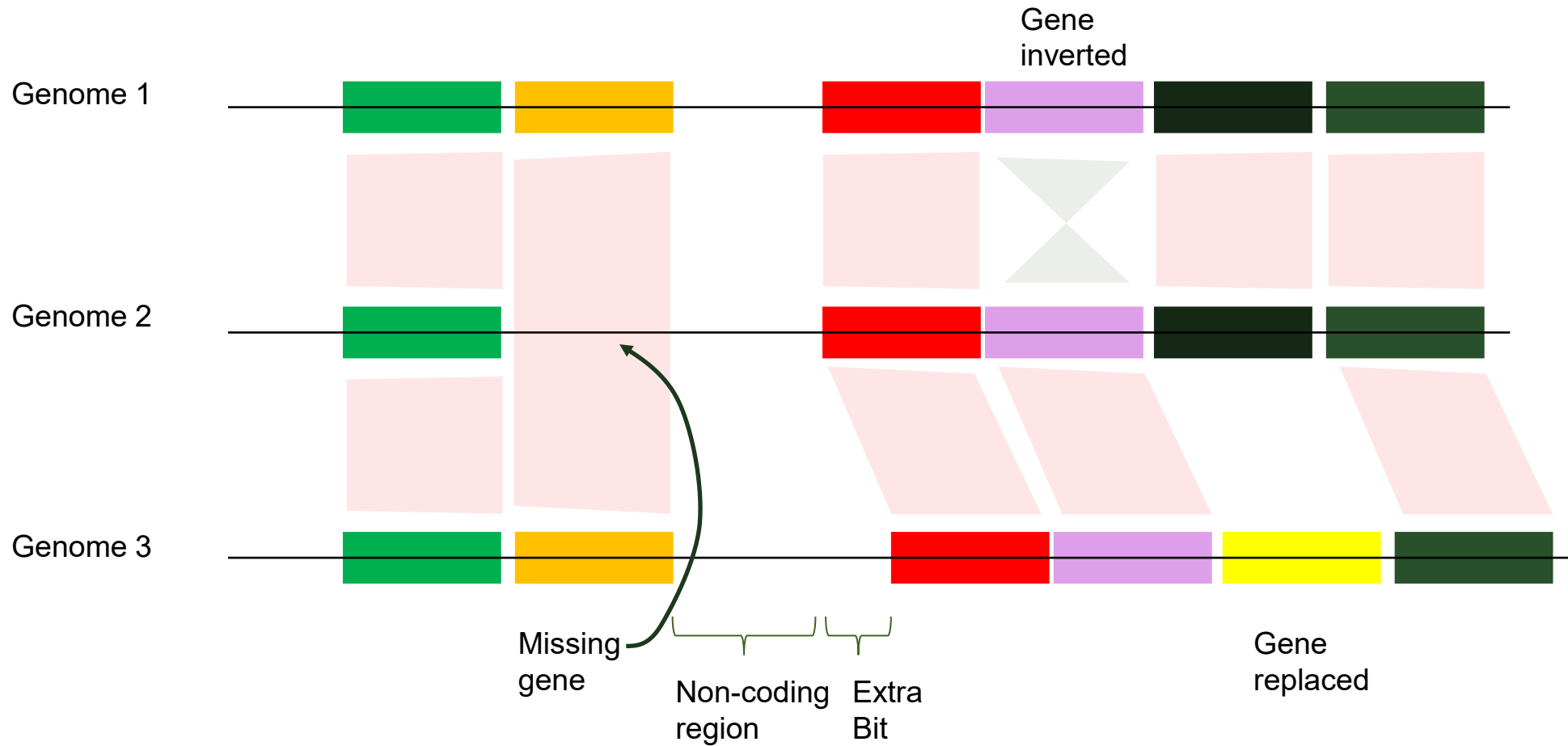
MSA is complex and easy to do badly!

Considering the bacterial pangenome



- For many species, especially those with an 'open pangenome'
 - Increased evolutionary distance = Fewer 'core' genes in common
 - Bacterial genomes that are very distant may have only a small number of core genes (e.g. 1-2000)
 - Bacterial genomes from the same ST or with a very recent common ancestor may have many more core genes between them (e.g. 3-4500)
 - This affects how we make an alignment

How do I make an alignment of bacterial genomes?



How do I make an alignment of bacterial genomes?

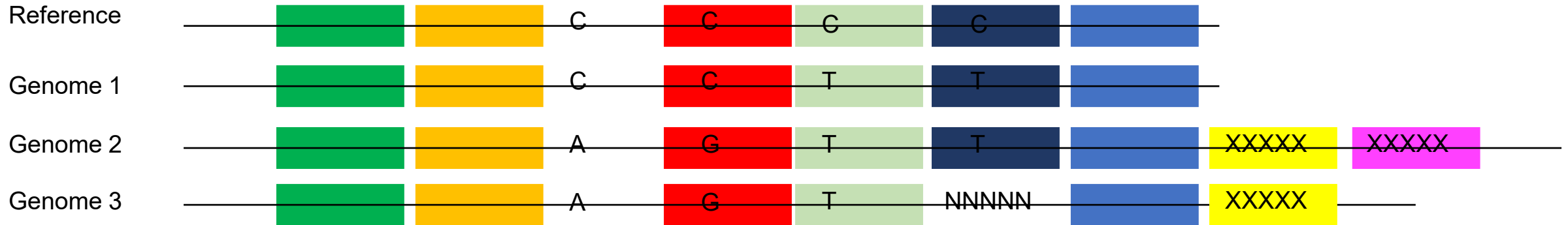
Bacterial genomes are messy

Picking the right approach for the question

Different methods

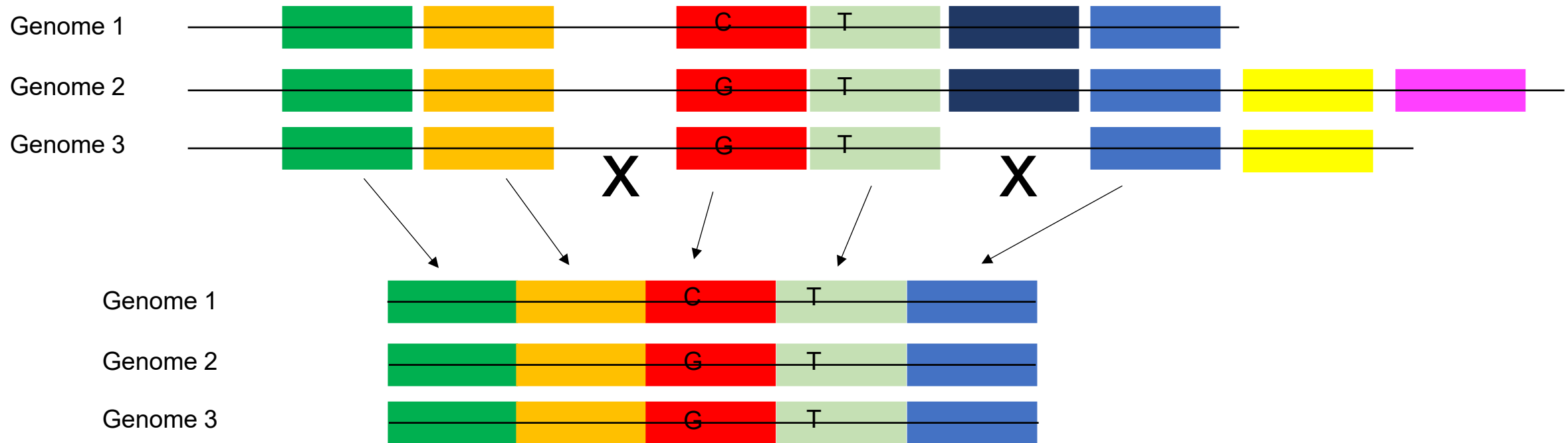
- Reference Mapping
- Core gene alignment
- Core genome alignment
- Graph based alignments

Reference Mapping



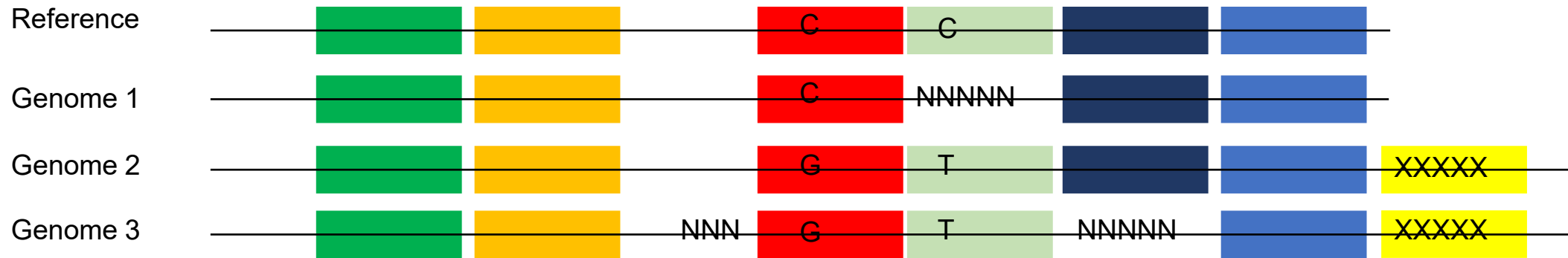
- Uses a single reference genome as a structure
 - Relates every other genome to that reference
- Can lead to biases:
 - Reference genes are missing from sample genomes
 - Misses genomic diversity absent from reference but present in sample
 - Good for identifying SNPs, bad for structural variation
- If samples are closely related to the reference (e.g. within lineage, outbreak), this is often the best approach
 - Often not appropriate for long term evolution (species level), especially highly diverse pathogens

Core Gene Alignment



- Uses a pangenome tool (e.g. roary, panaroo) to identify gene orthologs
- Keeps only 'core' genes present in all genomes
 - Discards 'accessory genes'
 - Discards non-coding regions
- Concatenates individual gene alignments together, all in +strand direction
 - Structural/positional information is lost

Core Genome Alignment



- Attempts to align *de novo* assemblies together (often guided by a reference)
 - E.g. manual Mummer alignments, Mauve, Parsnp.
- Can keep some non-coding regions
- Often still discards or masks regions not included in all genomes. May retain some positional information.

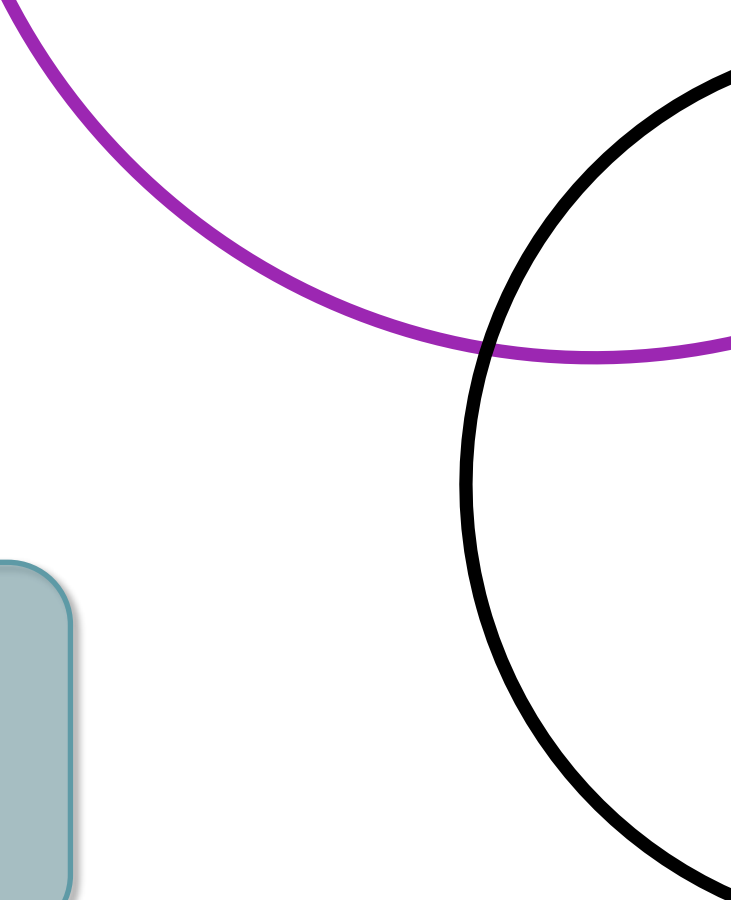
Pangenome Graphs

Builds a graph of pangenome, and infers SNPs between genomes that share part of the graph

E.g. Pandora

Very new and experimental

Computationally intensive



Construct phylogeny

How do we calculate a tree?

Why do we need computational methods?

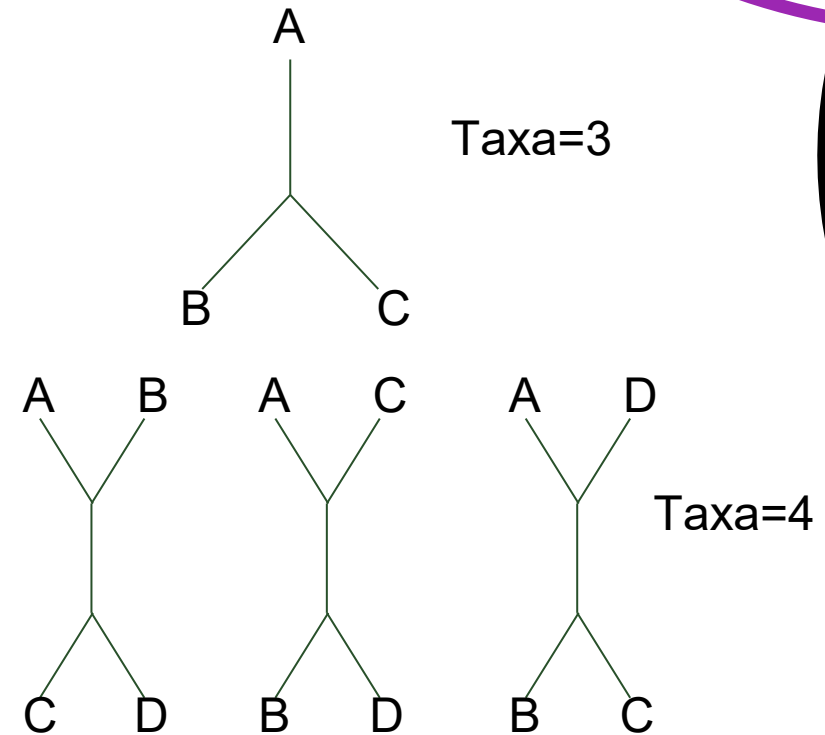
For any given dataset, there are many possible trees

Number of taxa	Number of unrooted trees
3	1
4	3
5	15
50	$\sim 2.84 \times 10^{74}$

Number of grains of sand on earth $\sim 7.5 \times 10^{18}$

Number of protons in universe $\sim 1 \times 10^{80}$

Therefore, need an optimized way of determining the optimal tree



How are simple trees constructed?

Given the data, we use algorithms to find the most probable tree (from many possible trees), using:

- An evolutionary model to determine...
- ...Genetic relatedness between taxa
- This can be computationally demanding

Construct phylogeny: choices

Method	Data used	Tree search	Evolutionary Model
Distance	Pairwise distance	Simple algorithm	Can be complex
Parsimony	All sites	Mainly hill climbing	Simple
Maximum likelihood	All sites	Hill climbing	Can be complex
Bayesian Methods	All sites (+ other info)	MCMC	Can be very complex

Phylogenetic Methods

Method	Data used	Tree search	Evolutionary Model
Distance	Pairwise distance	Simple algorithm	Can be complex

Position:

1

11

 21

Taxa A:

ATGCATATTT

AAGCATTATA

 ACTGATCATG

Taxa B:

ATG**G**ATAG**G**TT

TAGC**G**TTATA

 ACTG**C**TCATG

Differences:

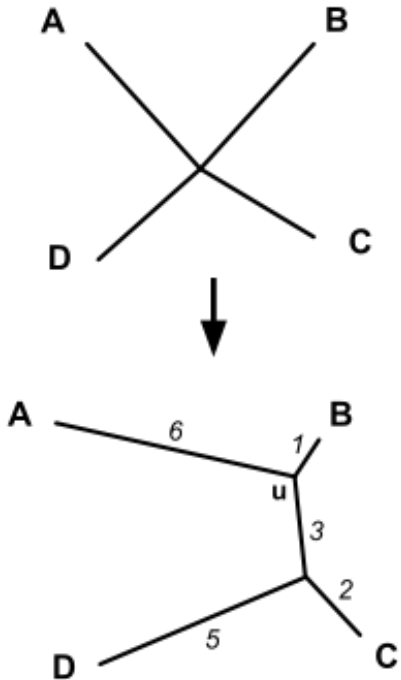
000**1**000**1**00

1000**1**00000

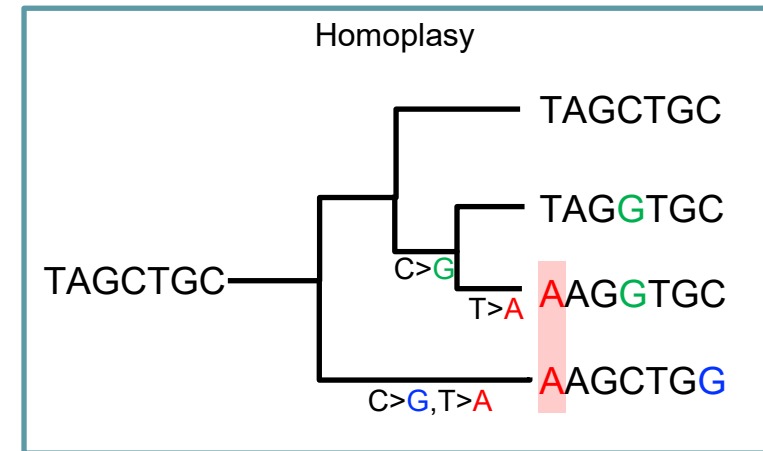
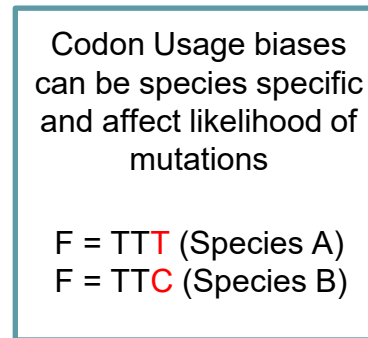
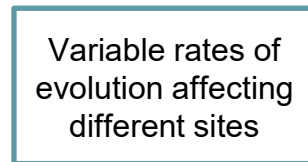
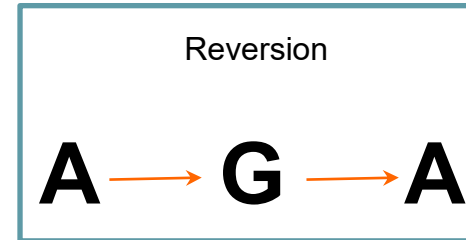
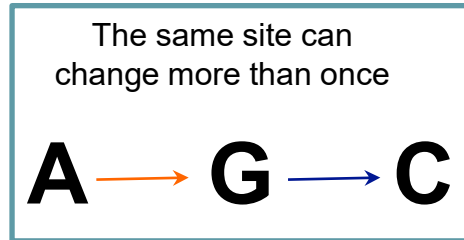
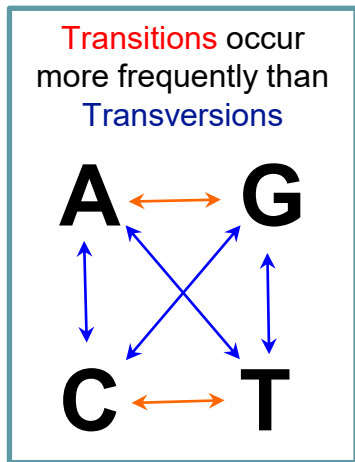
 0000**1**00000

Taxa A vs Taxa B = **5 (0.167)**

	A	B	C	D
A	0	7	11	14
B	7	0	6	9
C	11	6	0	7
D	14	9	7	0



But biology is more complicated...



- Models need to account for different substitution frequencies and biases
- Summarising sequence data as a distance matrix results in a loss of information that reduces accuracy

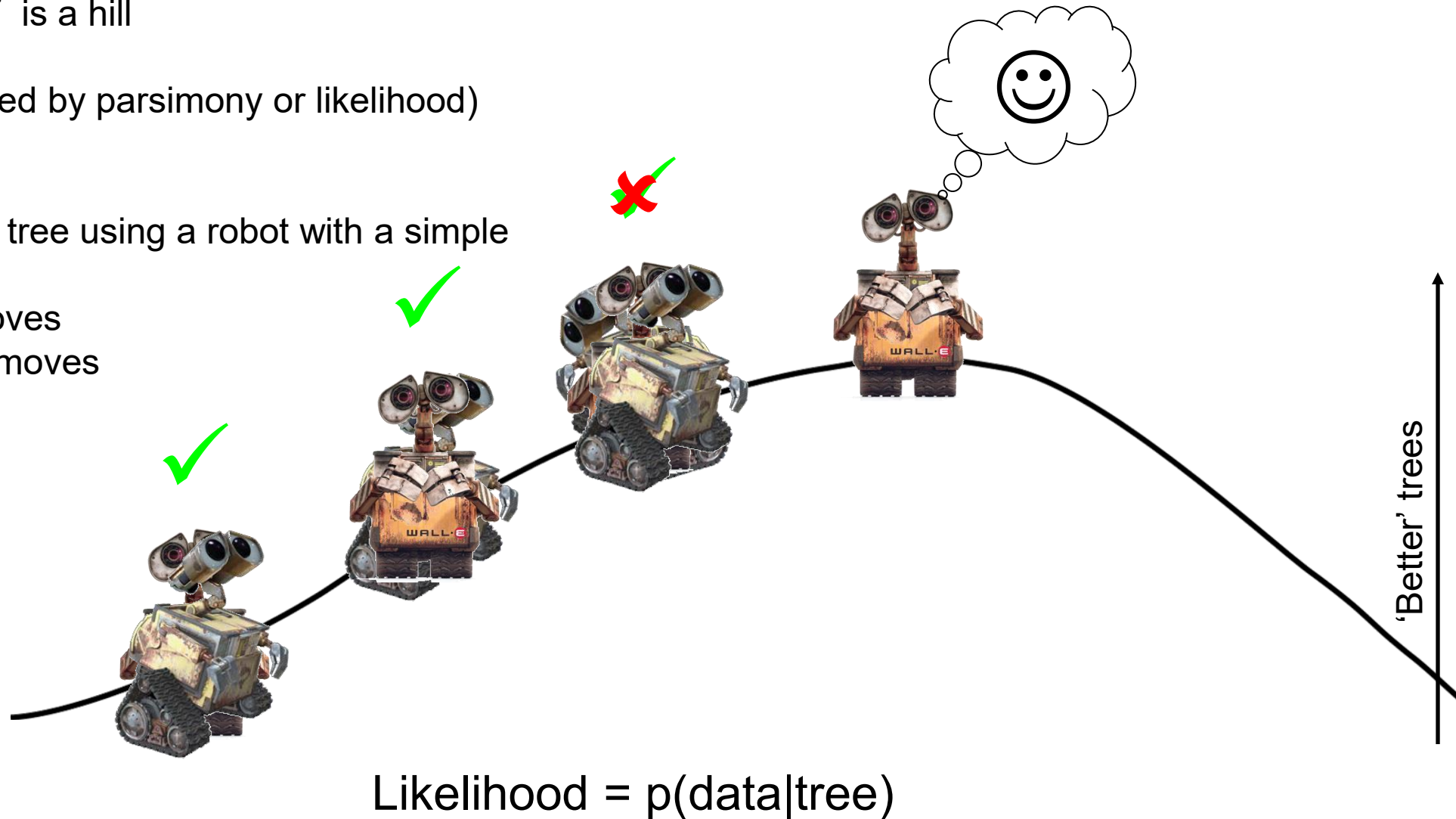
Phylogenetic Methods

Method	Data used	Tree search	Evolutionary Model
Distance	Pairwise distance	Simple algorithm	Can be complex
Parsimony	All sites	Mainly hill climbing	Simple
Maximum likelihood	All sites	Hill climbing	Can be complex
Bayesian Methods	All sites (+ other info)	MCMC	Can be very complex

$$\text{Likelihood} = p(\text{data}|\text{tree})$$

Finding the optimal tree – Hill Climbing

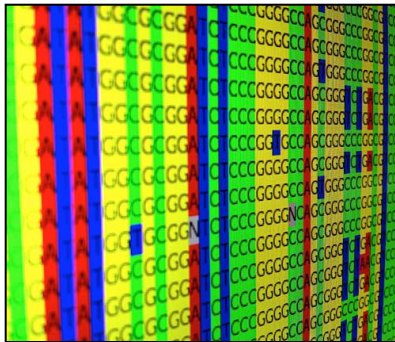
- Imagine tree 'space' is a hill
- Better trees (measured by parsimony or likelihood) are higher
- We can find the best tree using a robot with a simple program:
 - Accept uphill moves
 - Reject downhill moves



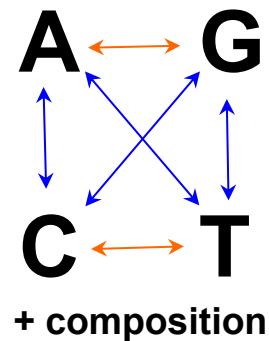
Construct phylogeny: Maximum likelihood

Maximum likelihood phylogenetic models maximise the probability of achieving ...

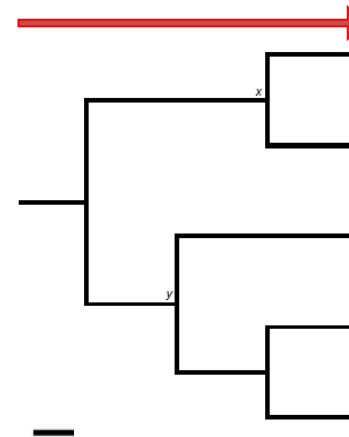
these data...



... if this happens...



... over this tree



NB: many more complex sequence evolution models exist to try and combat sources of molecular homoplasy that might result in the wrong tree including: substitutional saturation, *rate heterogeneity*, codon usage bias, *base composition bias*, secondary structure, *covariance*, rate convergence

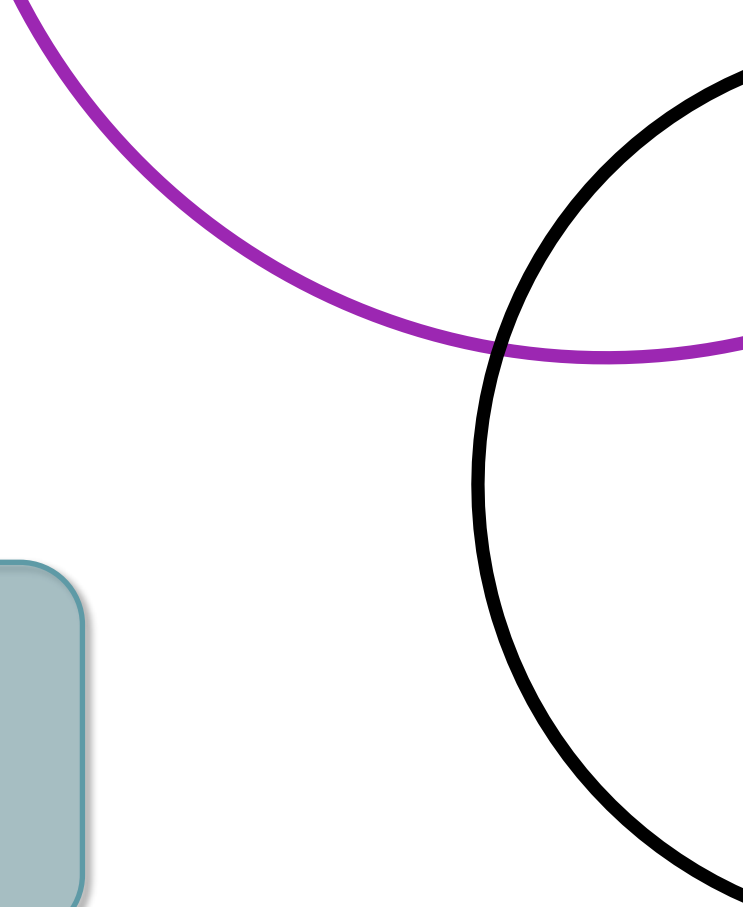
Maximum likelihood

Advantages:

- Can be highly accurate because considerable biological realism is introduced through the substitutional model. This allows various forms of homoplasy to be corrected for.
- Phylogenetic estimation within the likelihood framework provides a robust statistical context in which to evaluate specific hypotheses.
- A single tree is produced that is generally precise.

Disadvantages:

- The complexity of the estimation process means that it is slow and computationally demanding.
- The hill-climbing algorithm is susceptible to local optima and so does not guarantee to return the most optimal solution.



How trustworthy is my tree?

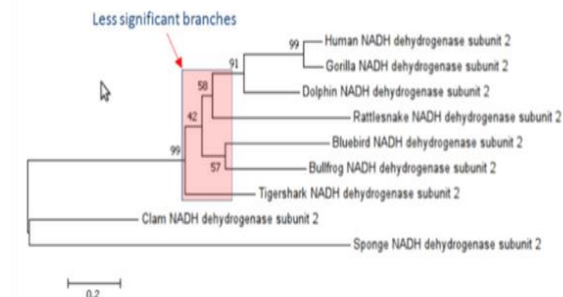
ML/Bayesian methods give overall metrics for tree, but individual nodes may be more or less reliable

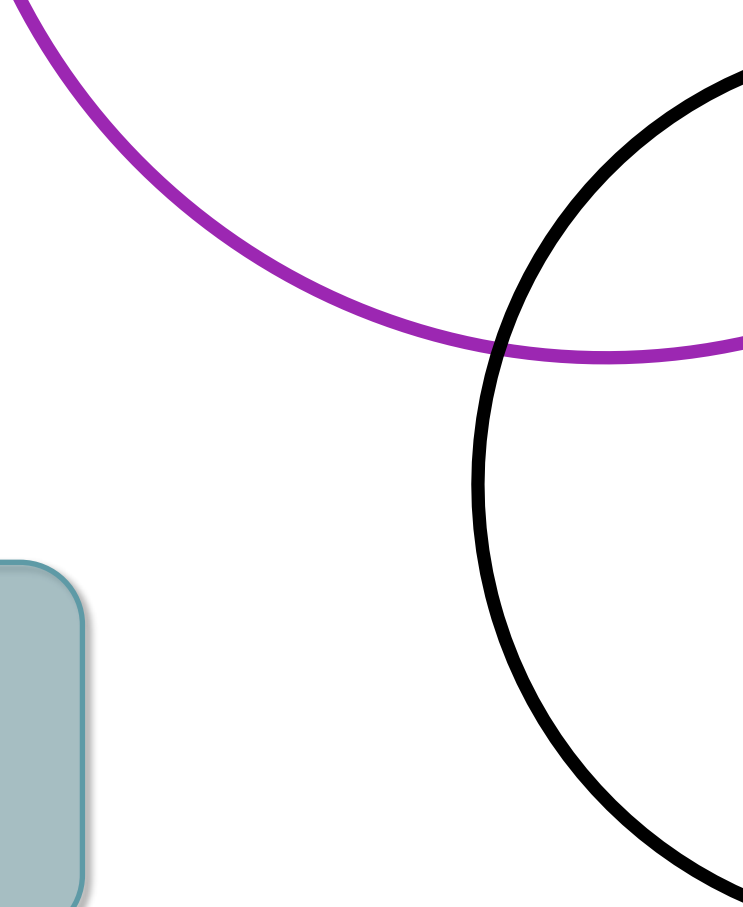
Confidence measures: Bootstrapping

- **Bootstrapping** is a way to produce a confidence measure in the topology relationships found in a phylogenetic analysis
- X number of **bootstraps** (resampled replicates) are created of your input data (MSA)
- To create a **bootstrap**, alignment columns are resampled at random. Each bootstrap is then reanalysed as per your phylogenetic construction algorithm
- The frequency of monophyletic clades (nodes) in bootstrap trees (**bootstrap proportions**) is then calculated e.g. 92/100 bootstraps
- These are commonly used as a measure of support for these branches and are represented as a number on each tree branch

Taxa	Characters								
	1	2	3	4	5	6	7	8	9
A	A	C	C	T	G	A	T	G	C
B	A	G	C	T	G	G	T	T	C
C	A	G	C	A	G	A	T	G	G
D	T	C	C	T	C	G	T	G	C
E	T	C	T	T	A	A	T	G	C

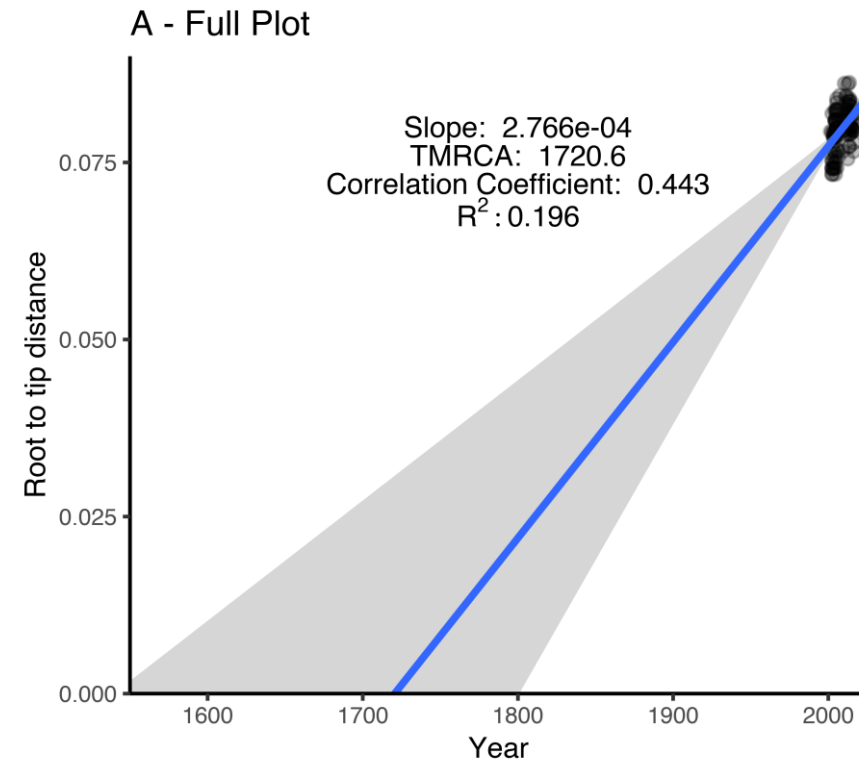
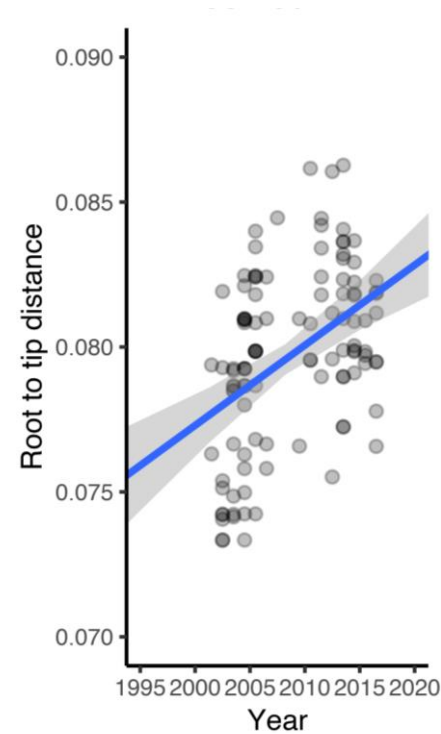
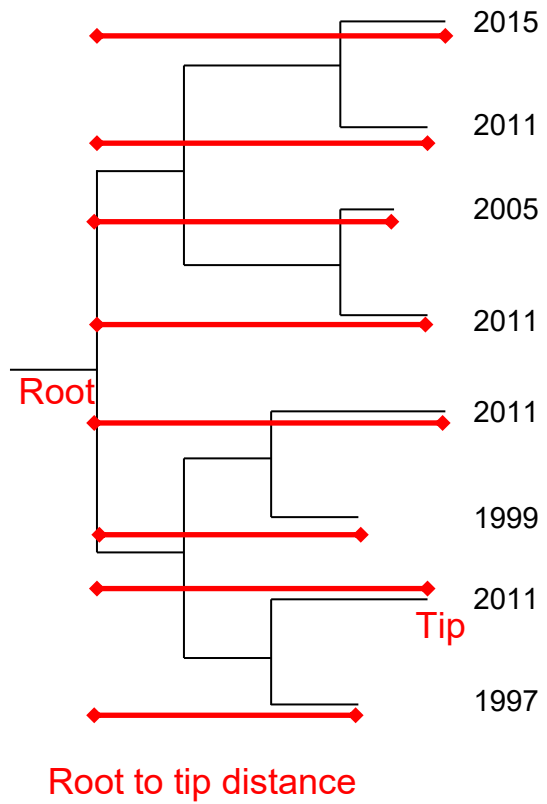
	Characters								
Taxa	2	5	9	2	7	7	2	1	6
A	C	G	C	C	T	T	C	A	A
B	G	G	C	G	T	T	G	A	G
C	G	G	G	G	T	T	G	A	A
D	C	C	C	C	T	T	C	T	G
E	C	A	C	C	T	T	C	T	A

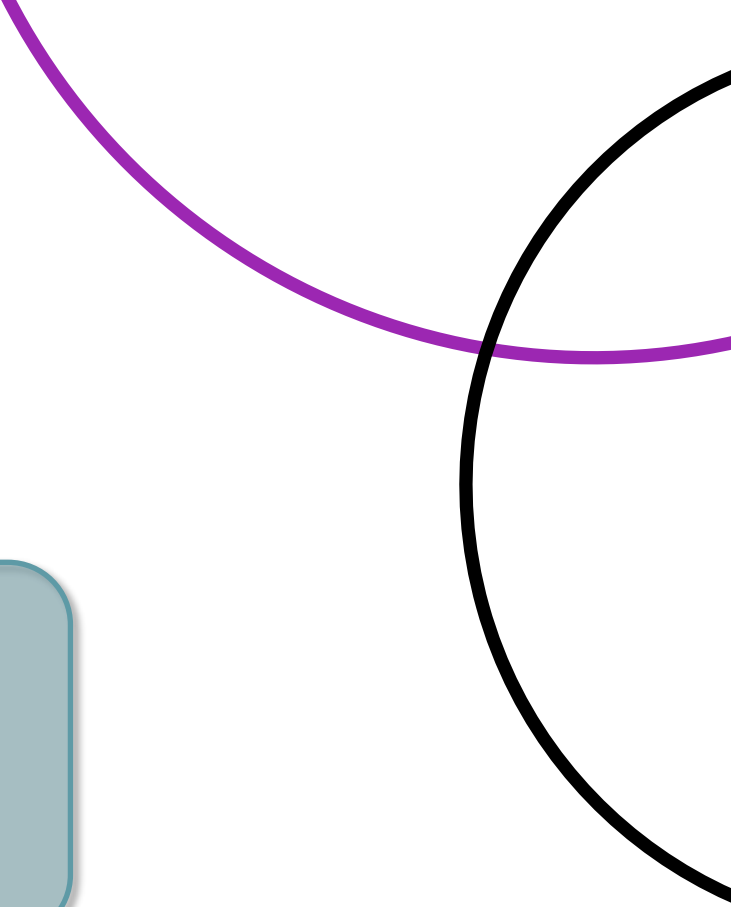




How do we infer timelines?

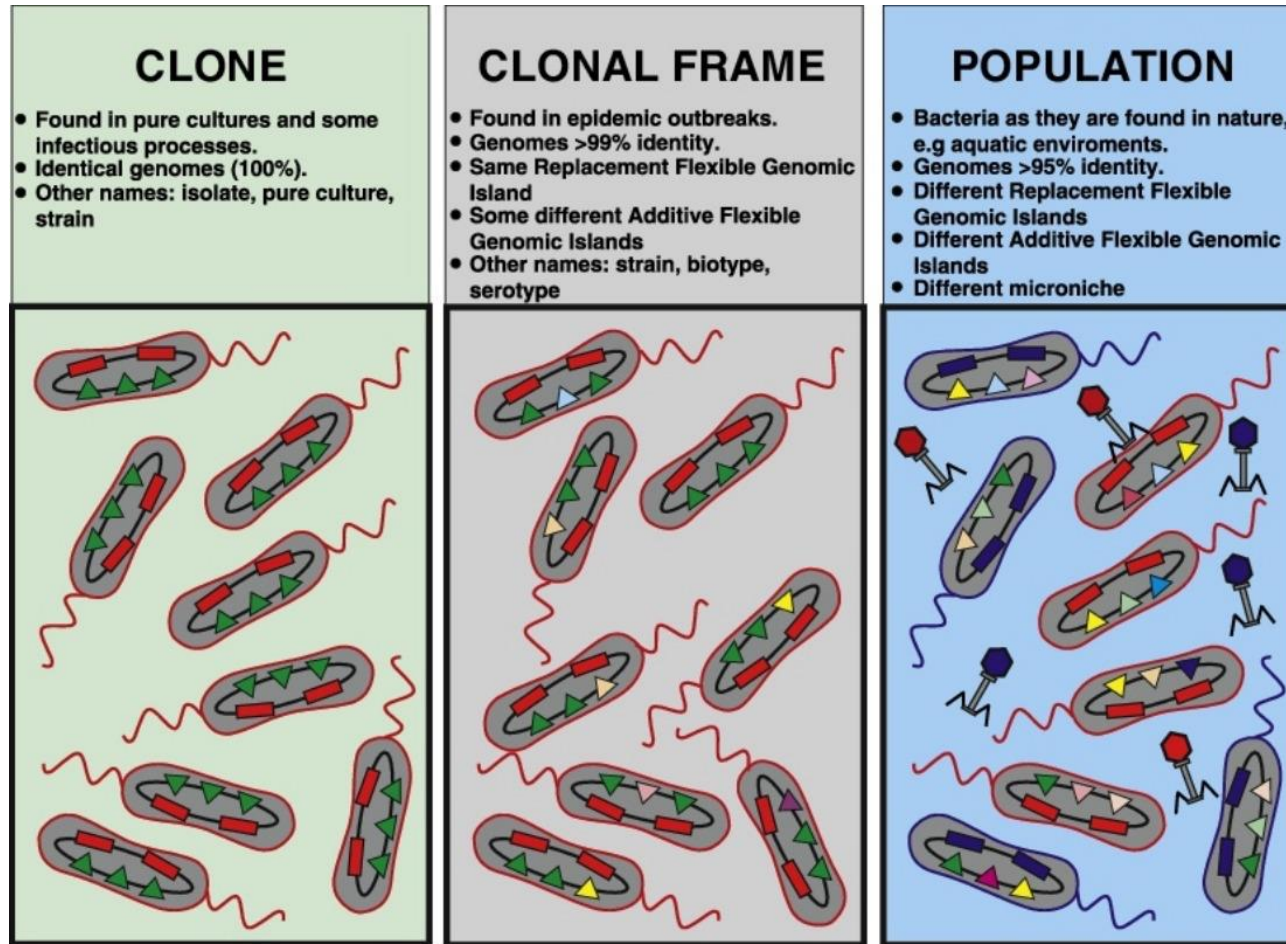
Temporal inference





Phylogenetic considerations for Bacteria

Bacterial genomes can be related in different ways

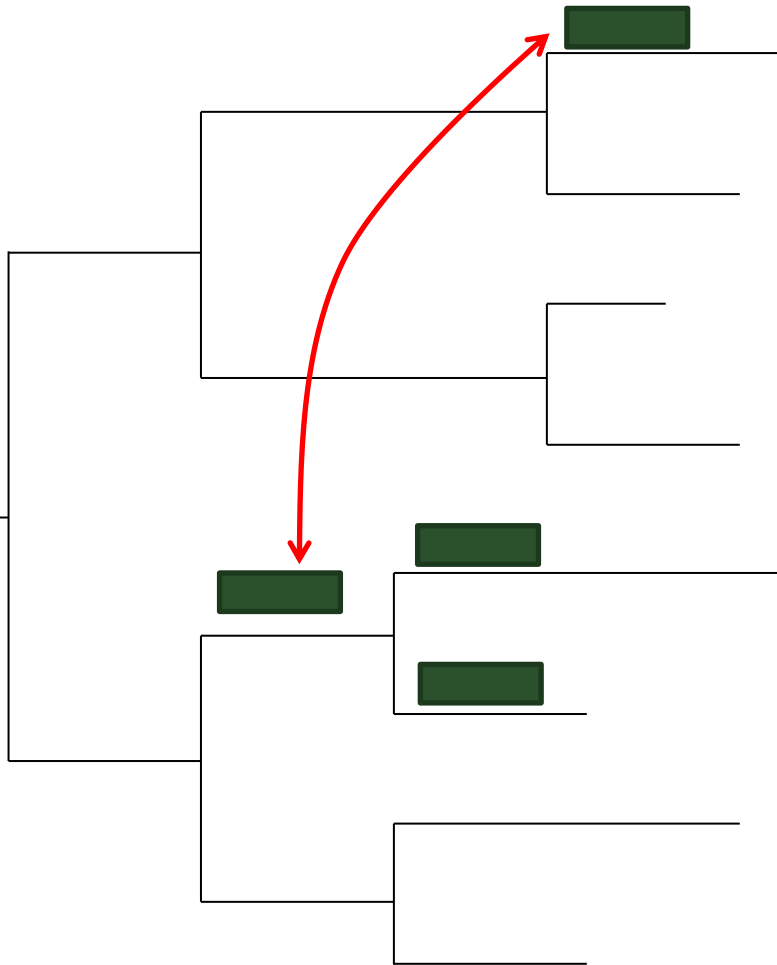


□ Replacement Flexible Genomic Islands (coding for exposed structures)

△ Additive Flexible Genomic Islands (coding for microniche adaptation)

- Phylogenetics relies on the concept of vertical descent and accumulation of variants (common ancestor)
- In many bacteria, much of the genome is fixed in a single chromosome. This inherited backbone is called the clonal frame
- Variants in the clonal frame are accumulated through drift and selection in a stepwise manner

Recombination is important



Recombination and Horizontal Gene Transfer are contrary to vertical ancestry, because :

- Large blocks of alleles (SNPs, INDELS, Genes) are transmitted together as a single event
- These may be absent from common ancestors but shared with other parts of the tree
- This can distort the topology and branch lengths of a phylogeny, giving false results

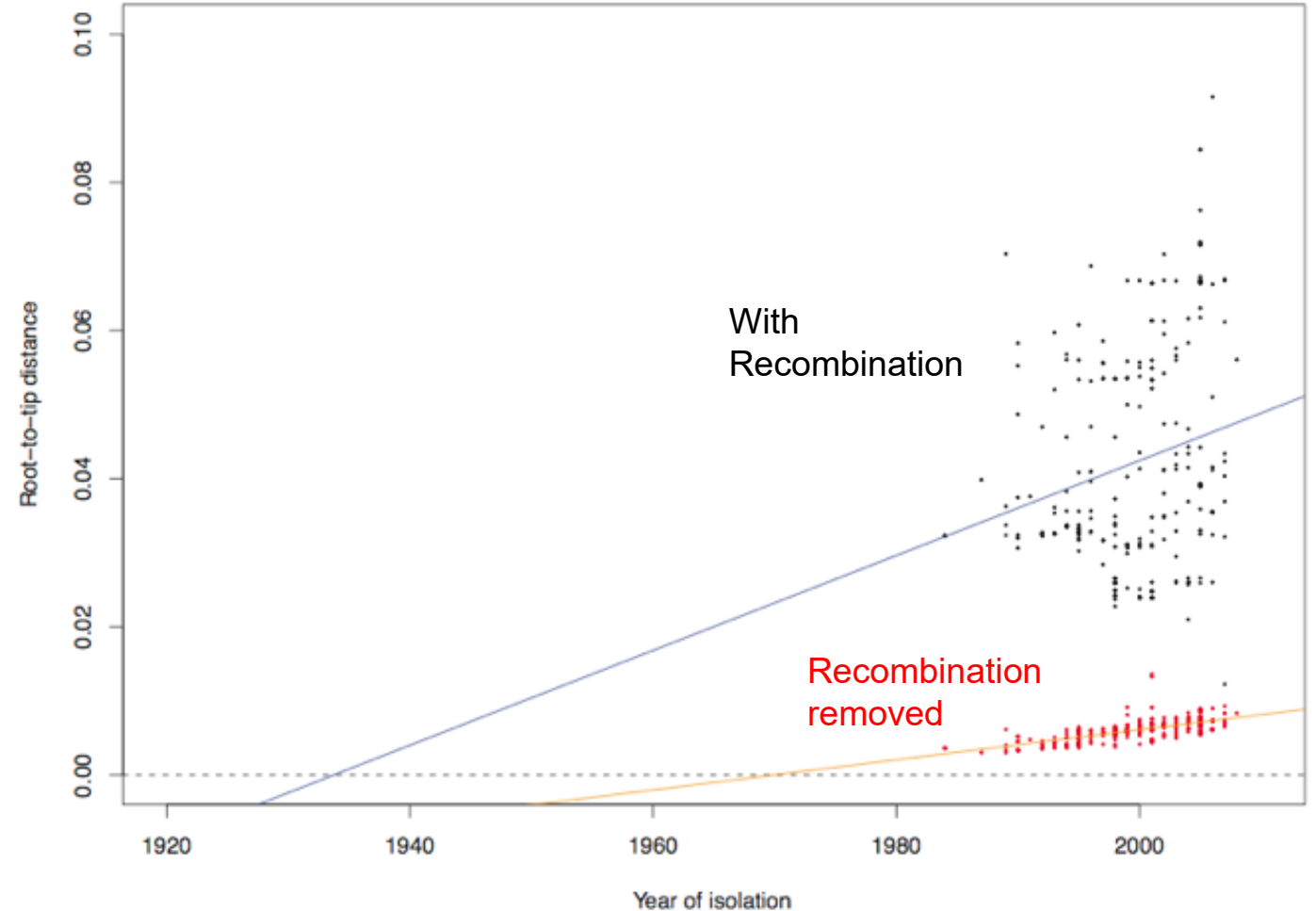
Recombination is important

For phylogenetics it is important to remove or account for recombination

Analysis should be performed only on the clonal frame NOT on the recombining sites.

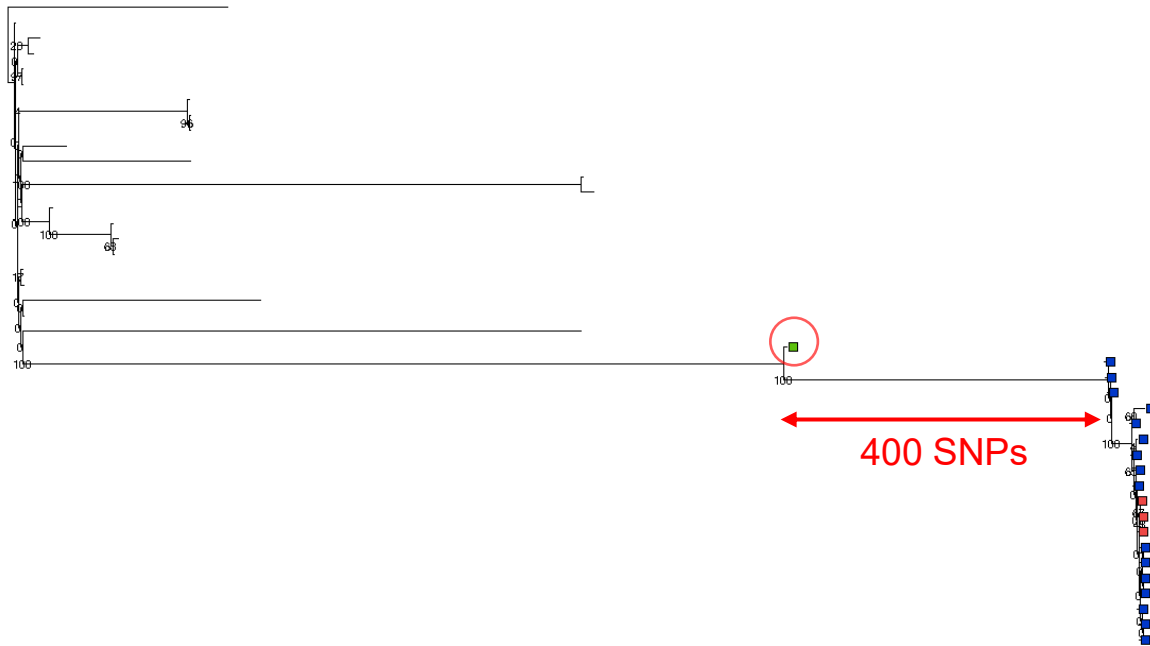
Recombination can be screened for using a number of tools

- e.g. Gubbins, ClonalFrameML

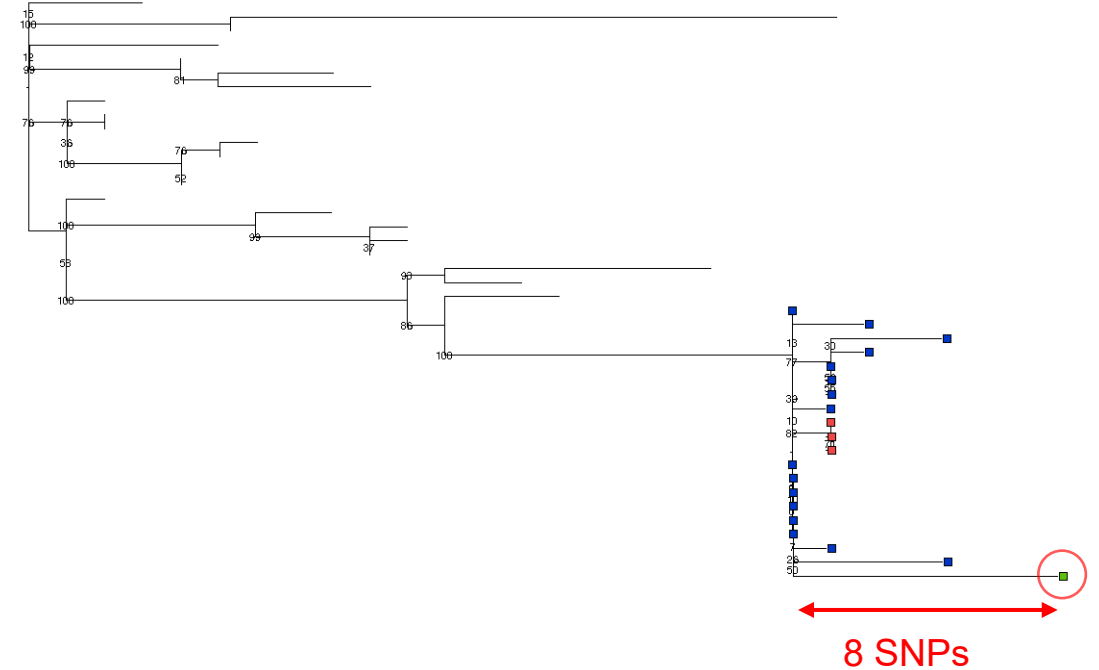


Recombination is important

Raw Sequence Alignment



Recombination Removed



Recombination is important

Caution

- Removing/masking recombination may hide important diversity that can discriminate genetically close samples.
- May make samples appear closer to one another than they really are
- This can become a confounder in genomic epidemiology
- Masking recombination may not always be necessary to address a question

Methods like Gubbins are intended for 'within-species/lineage' analysis

- For very long-term evolutionary timescales in many bacteria, phylogenetic signal and recombination become indistinguishable
 - Therefore unsuitable for genus level analyses
- Not suitable for use with 'core gene alignment' methods, because positional synteny is lost.

Always worth considering why you are removing recombination

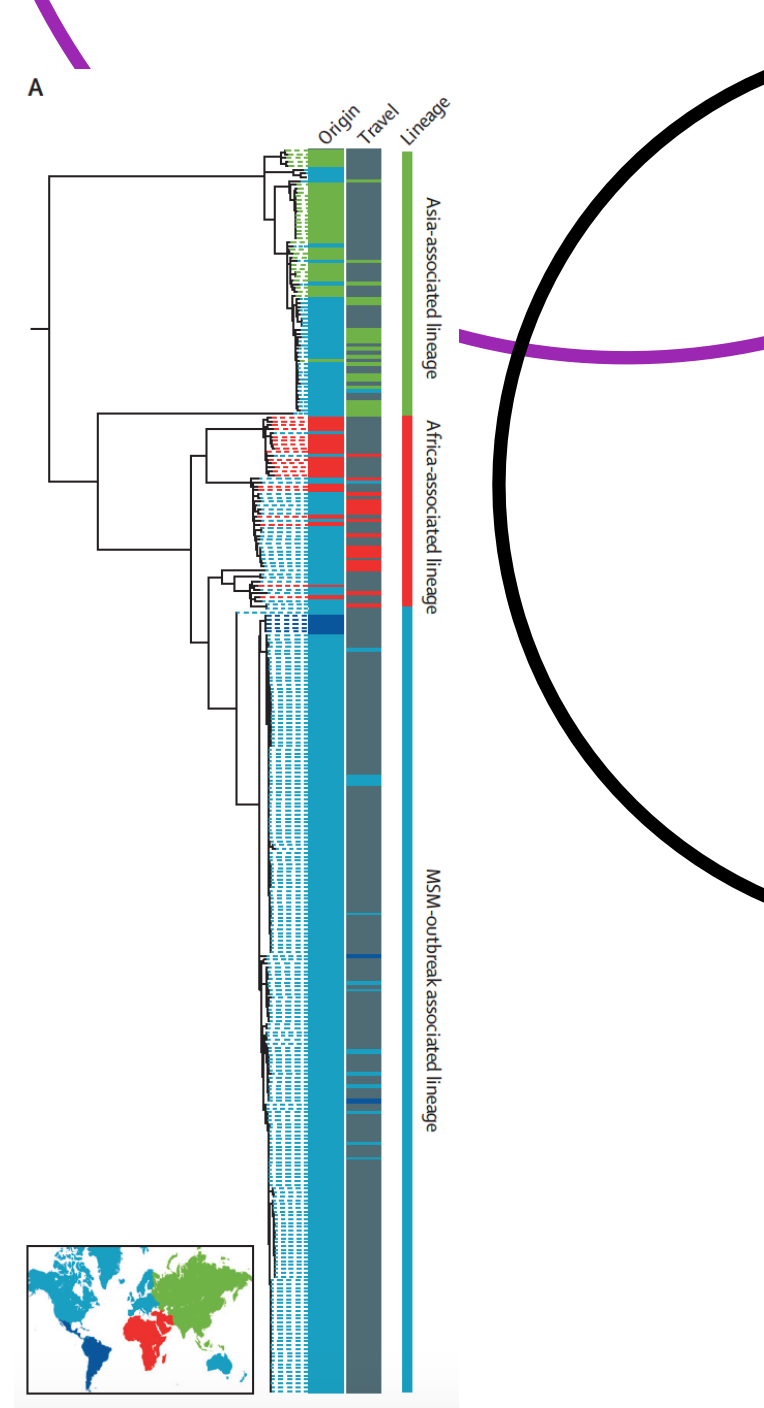
Recombination is important

Some bacteria may recombine so much that the Clonal Frame is rapidly lost, e.g. *H. pylori*

- In these cases, phylogeny is usually inappropriate and other methods must be used

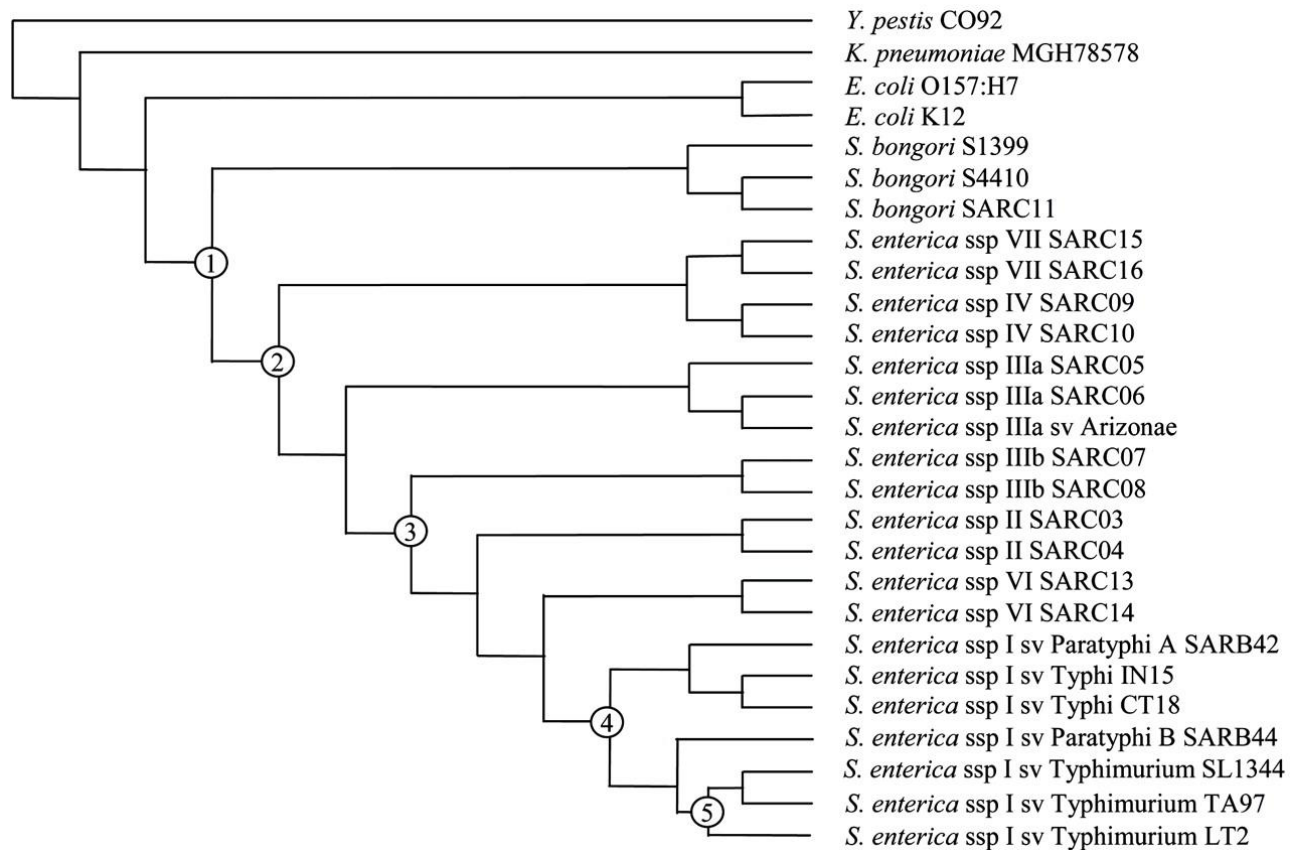
A Tree is **Not** the Endpoint

The tree provides an evolutionary framework for investigating phenotypes, genotypes, metadata, etc.



Why is Phylogenetics Useful?

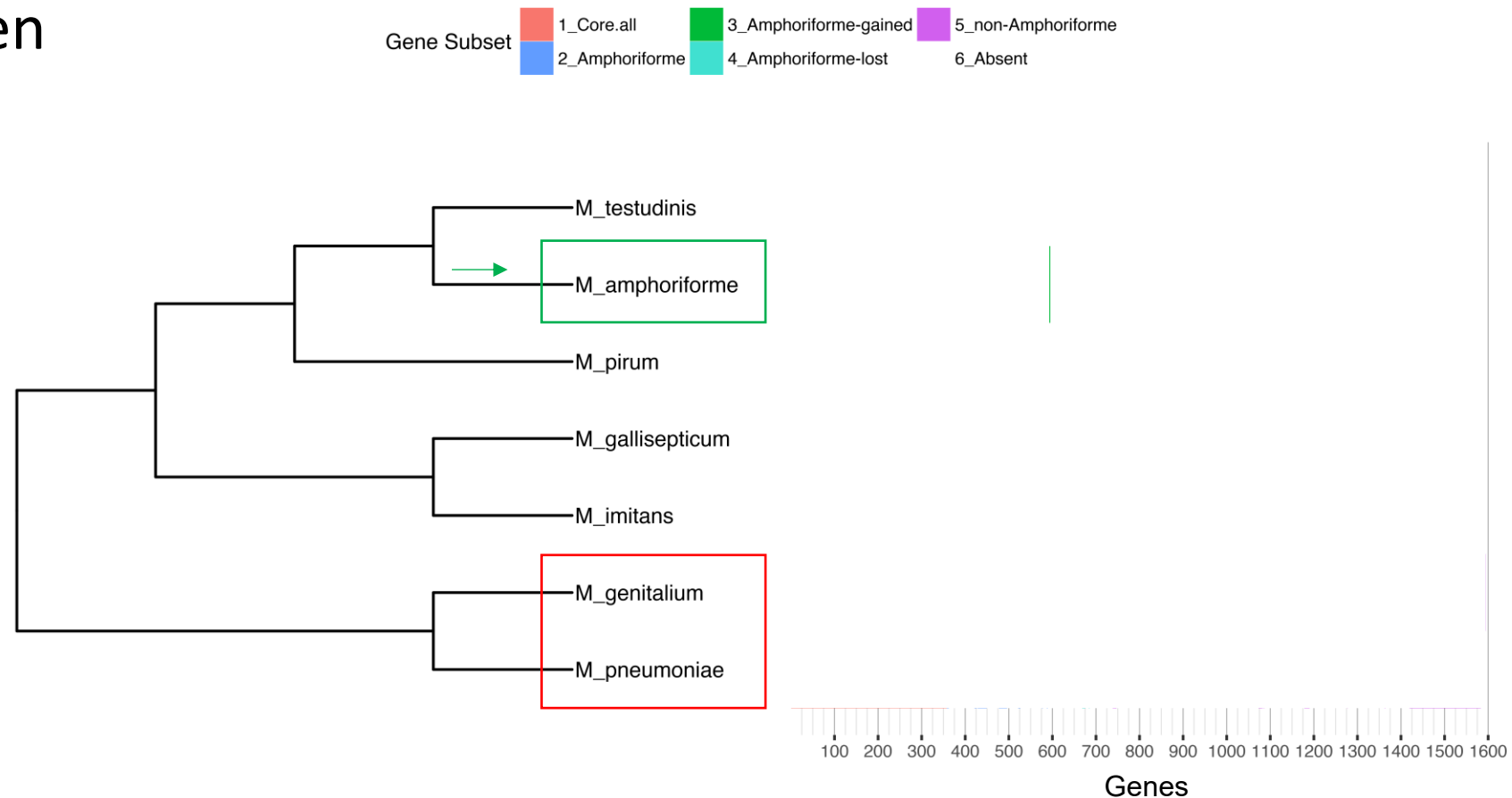
Show how pathogens are related to each other



Why is Phylogenetics Useful?

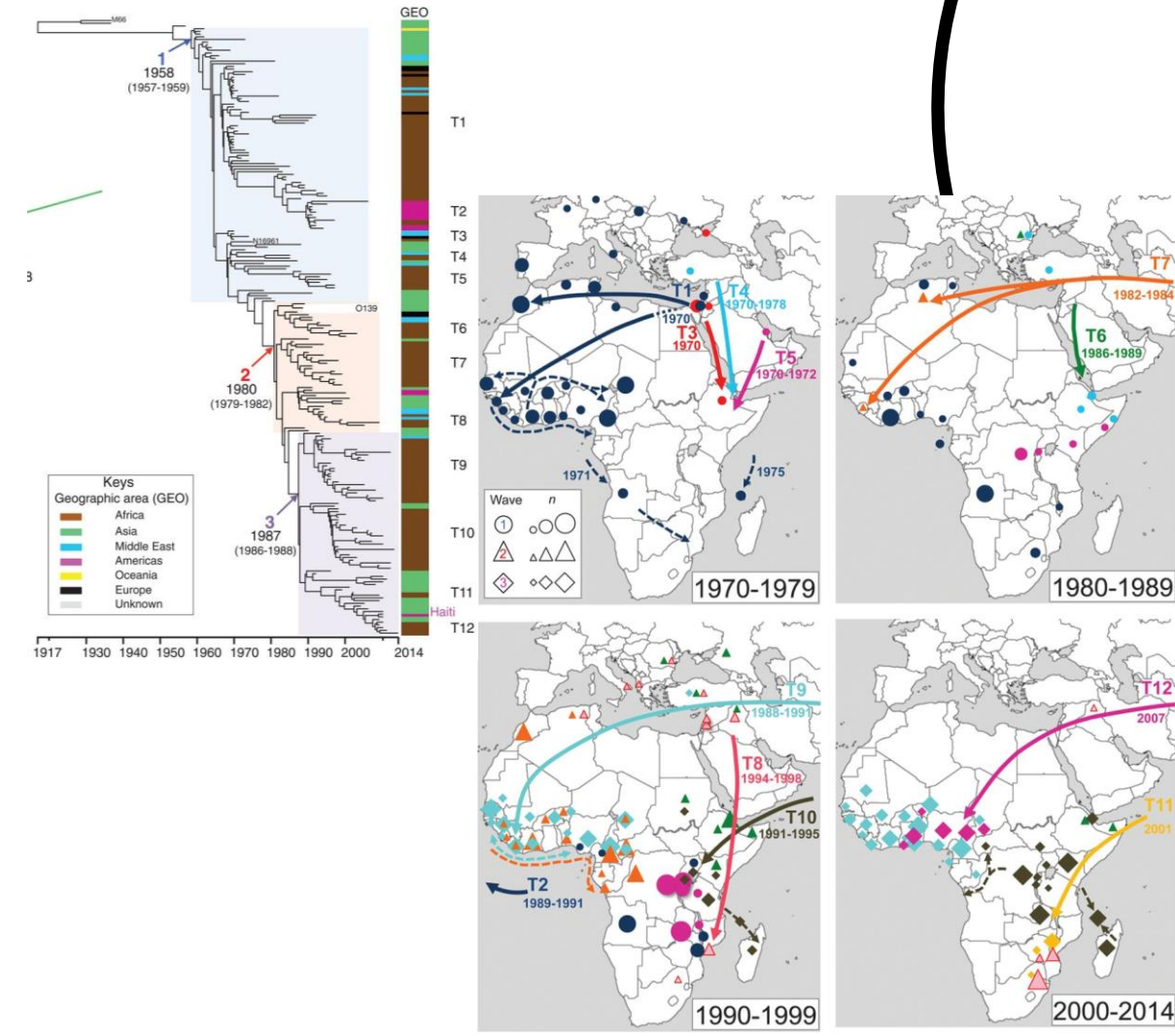
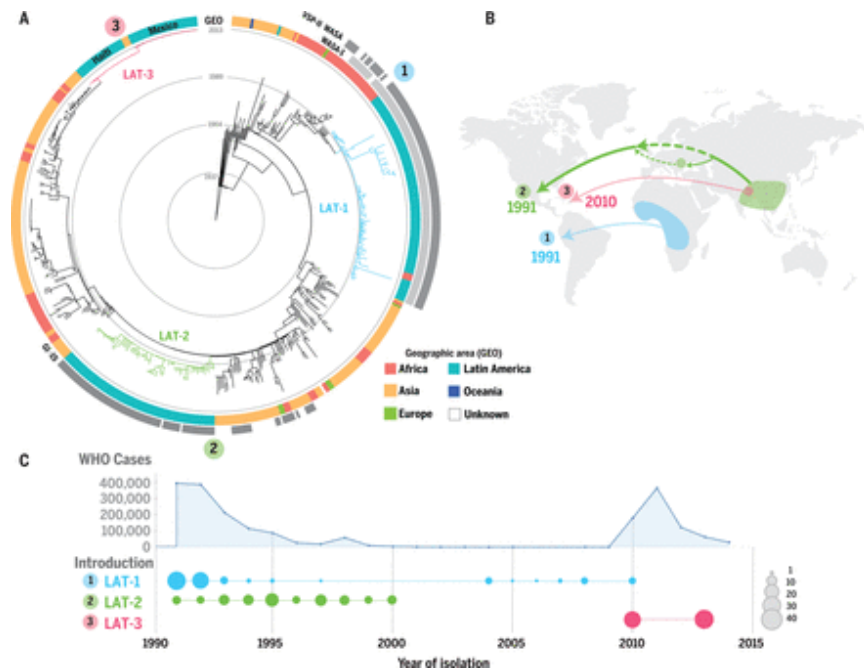
- Gain insights into pathogen evolution

- Pathogen evolution and host adaptation
- Genes retained and lost
- Adaptation to lifestyle?



Why is Phylogenetics Useful?

Study of international disease and patient-to-patient transmission



Why is Phylogenetics Useful?

Real-time tracking and reconstruction of a global pandemic



WHO WE ARE HOW WE WORK UPDATES DATA PROTOCOLS STUDIES CONTACT

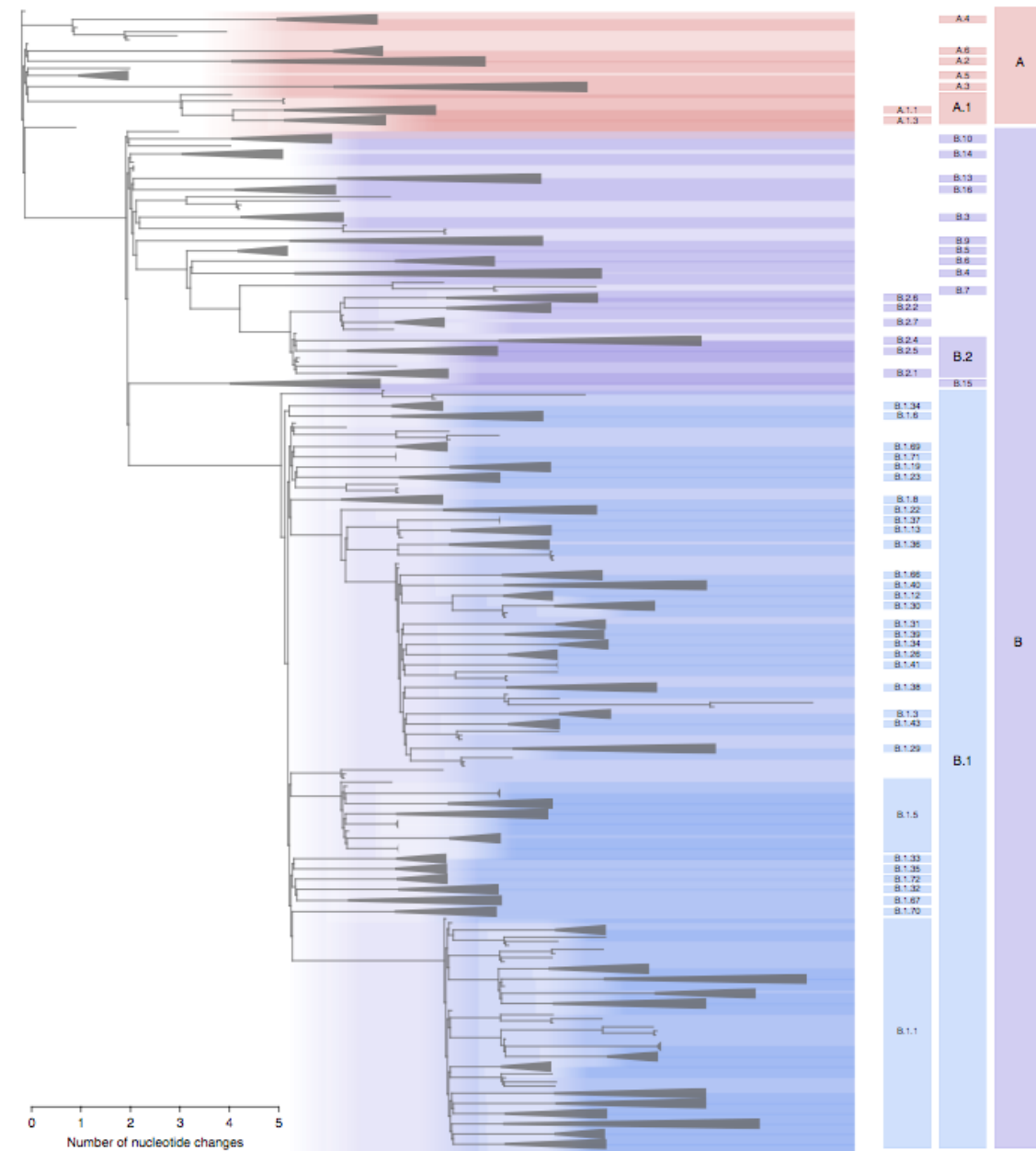
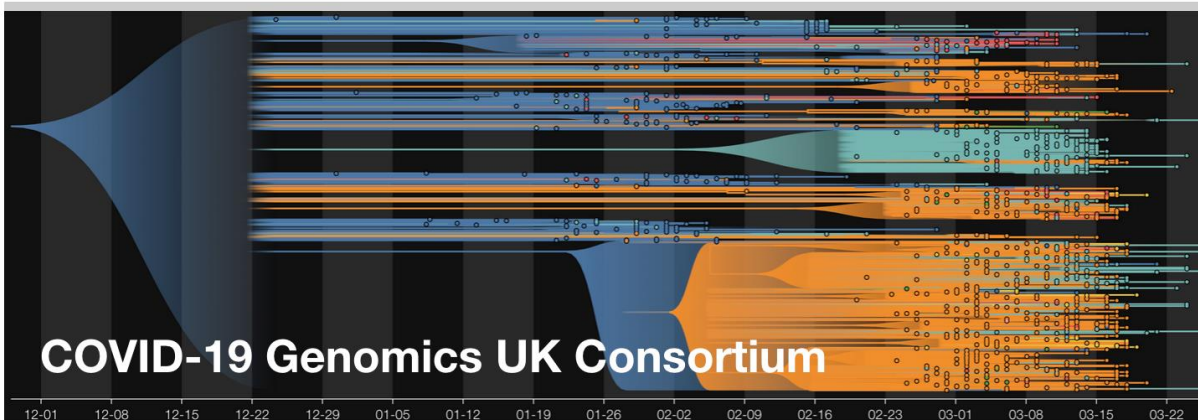


Fig.1 | Maximum likelihood phylogeny of globally sampled sequences of SARS-CoV-2 downloaded from the GISAID database on 18 May 2020. Five representative genomes are included from each of the defined lineages. The largest lineages defined by our proposed nomenclature system are highlighted with coloured areas and labelled on the right. The remaining lineages defined by the nomenclature system are denoted by triangles. The scale bar represents the number of nucleotide changes within the coding region of the genome.

Bacterial_Genomic_Surveillance_2026 / course_modules_2026 / Mapping_and_Phylogenetics / Mapping+Phylo.md

Preview Code Blame 901 lines (583 loc) · 30.5 KB Raw

Phylogenetics

Now that we have a clean multiple sequence alignment, we are now going to use `IQ-TREE` to build a maximum likelihood phylogeny.

Make a SNP-only alignment using `snp-sites`

Calculating a phylogeny on whole genome sequences can be very time consuming. We can speed this up by only using the variable sites (SNPs). However, we need to be aware that only including variable sites can affect the evolutionary rate estimates made by phylogenetics software - therefore, we need to account for the sites we remove in our analysis.

We will use `snp-sites` (<https://github.com/sanger-pathogens/snp-sites>) to do this. You can view the options for `snp-sites`

```
snp-sites
```

```
(base) vboxuser@Ireland2026:~/Module_4_Mapping_Phylogeny$ snp-sites
Usage: snp-sites [-rmvpcbhV] [-o output_filename] <file>
This program finds snp sites from a multi FASTA alignment file.
-r      output internal pseudo reference sequence
-m      output a multi fasta alignment file (default)
-V      output a VCF file
```

Thank you!



Dr. Daryl Domman



Dr. Mat Baele

Make a SNP-only alignment (snp-sites)

Calculate a phylogenetic tree from the SNPs (IQ-TREE2)

Visualise a phylogenetic tree (FigTree)

QC

Visualise a tree using Microreact



Account for recombination (gubbins)

Root a phylogeny

Visualise recombination (phandango)



Cluster genomes (fastBAPS)

Visualise Tree using Microreact

